**NOT MEASUREMENT SENSITIVE**

MIL-STD-1812A
OCTOBER 9, 2020

SUPERSEDING
MIL-HDBK-1812
FEBRUARY 14, 1997

# DEPARTMENT OF DEFEENSE
# STANDARD PRACTICE

# AERONAUTICAL AND SUPPORT EQUIPMENT TYPE DESIGNATION SYSTEM

**Reinstated after 9 OCT 2020 and may be used for new and existing designs and acquisitions.**

AMSC 10195 AREA SESS

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## FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense (DoD).

2. This standard is mandatory for use by the Department of the Air Force for designating digital imaging equipment, aeronautical equipment, support equipment, munitions and aerospace vehicle engine types. The designation of electronic materials is accomplished in association with MIL-STD-196 and the Joint Electronics Type Designation Automated System (JETDAS).

3. This standard establishes the requirements for obtaining official assignment of nomenclature and conformation of existing nomenclature for digital imaging equipment, aeronautical equipment, support equipment, munitions and aerospace vehicle engines acquired by the Department of Defense.

4. This standard has been created to consolidate documentation concerning type designation assignment and to standardize procedures for obtaining nomenclature. Contract End Item Numbers and Configuration End Item Numbers are not replacements for official military systems of type designations used throughout the Department of Defense.

5. The data contained herein supersedes data formerly contained in MIL-STD-155, MIL-STD-815, MIL-STD-842(USAF), MIL-STD-875, MIL-STD-879, MIL-STD-1557, and MIL-N-7513(USAF).

6. Comments, suggestions, or questions should be addressed to Engineering Standards, AFLCMC/EZSS, 2145 Monahan Way, Bldg. 28, Wright-Patterson AFB, OH 45433-7017. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at https://assist.dla.mil.



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# 1. SCOPE

## 1.1 Scope.

The purpose of this standard is to provide type designation systems and procedures for use by the Department of Defense (DoD) to standardize the identification of digital imaging equipment, aeronautical equipment, support equipment, munitions and aerospace vehicle engines. In addition, this standard defines the method for obtaining the nomenclature and type designations of these items

## 1.2 Objectives.

The type designation system is flexible and broad enough in scope to cover present types of equipment and equipment to be developed in the future; yet, be stable enough to permit the assigned type designation to continue to identify the equipment throughout its entire life. Further, these systems are designed to accomplish the following:

a. Simplify the identification of equipment and provide a means of readily identifying items designated;
b. Define and identify similar items to ensure that any significant differences between the user's viewpoint and that of the item's designed use and purpose should be readily distinguishable;
c. Provide a standardized means of identifying items on nameplates, in specifications, drawings, and technical data;
d. Provide an indication of interchangeability and substitutability;
e. Provide an unclassified means of identifying equipment in correspondence and other communications; and
f. Provide a means to disseminate unclassified technical data to activities within the departments and agencies of the DoD that justify a need.

# 2. APPLICABLE DOCUMENTS

## 2.1 General.

The documents listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

## 2.2 Government documents.

### 2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

**DEPARTMENT OF DEFENSE STANDARDS**

MIL-STD-196 Joint Electronics Type Designation Automated System



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**DEPARTMENT OF DEFENSE HANDBOOKS**

| | |
|---|---|
| MIL-HDBK-505 | Definitions Of Item Levels, Item Exchangeability, Models, and Related Terms |

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

**2.2.2 Other Government documents, drawings, and publications.**
The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

**DEPARTMENT OF DEFENSE**

| | |
|---|---|
| DoD 5220.22M | National Industrial Security Program Operating Manual |
| DoDM 5200.01, Volume 2 | DoD Information Security Program: Protection of Classified Information |

(Copies of these documents are available online at http://www.dtic.mil/whs/directives.)

**DEFENSE LOGISTICS INFORMATION SERVICE**

| | |
|---|---|
| H4 | Federal Supply Code for Manufacturers |
| H6 | Federal Item Name Directory |

(Copies of these documents are available online at http://www.logisticsinformationservice.dla.mil/hseries.aspx.)

**2.3 Non-Government publications.**
The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

**ASME INTERNATIONAL**

| | |
|---|---|
| ASME Y14.38 | Abbreviations and Acronyms for Use on Drawings and Related Documents |

(Copies of this document are available online at http://www.asme.org/.)

**SAE INTERNATIONAL**

| | |
|---|---|
| SAE EIA-649 | Configuration Management Standard (DoD adopted) |
| SAE EIA-649-1 | Configuration Management Requirements for Defense Contracts (DoD adopted) |

(Copies of these documents are available online at http://www.sae.org/.)



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### 2.4 Order of precedence.
Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. DEFINITIONS

### 3.1 Definitions.
For the purpose of this standard the following definitions apply:

#### 3.1.1 Aerospace vehicle.
Any air vehicle which qualifies for assignment of a mission-design-series designation per AFI 16-401, i.e., aircraft, missile, rocket, etc.

#### 3.1.2 Basic assignment.
A type designation assigned to identify the general design of the equipment, group, or unit. It consists of a type designation with an empty parentheses directly following the model number.

#### 3.1.3 Complement item.
Items or groups that make up the equipment or group to be assigned nomenclature. This item is expected to be part of the equipment or group and must be issued automatically with the equipment or group of which it is a part. All listed complement data must be unit level or above.

#### 3.1.4 Definitive equipment, groups, and units.
Definitive equipment, groups, and units should be those having a specific assemblage as defined in MIL-HDBK-505.

#### 3.1.5 Development activity.
Activity responsible for the development of an item.

#### 3.1.6 Development assignment.
A type designation assigned by the development activity to identify a development equipment, group, or unit after first obtaining a basic assignment from the DoDCP. It consists of the basic assignment and the appropriate development organization-indicator followed by a numeral.

#### 3.1.7 Electrical interchangeability.
The new item's capability of operation being equal to the old item without requiring any modifications to the existing power facilities, change to, or rewiring of connectors, etc.

#### 3.1.8 Equipment.
Air Force-Navy equipment is defined as a set, central, or system level of hardware, which requires an A/ level of type designation (see Figure 2).



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**3.1.9 Functional interchangeability.**
The new item's capability of performing, without additional assistance, all the operational capabilities covered by the previous item.

**3.1.10 Grandfathering.**
Grandfathering is the process of allowing the use of a non-approved item name to for a specific nomenclature that was completed during the manual Aeronautical Support & Equipment Type Designation System (ASETDS).

**3.1.11 Item.**
For purposes of this standard, an item is any level of hardware eligible for assignment of nomenclature.

**3.1.12 Item name.**
Names assigned by the activity authorized within the department for assignment of approved item names. Approved item names will typically be assigned by using the H6, the Federal Item Name Directory. Grandfathering is an example exception to this requirement.

**3.1.13 Maintenance (repair) parts interchangeability.**
The capability of installing and operating a maintenance part in a new item in lieu of a like part in an old item without the use of additional tools or modifications to the existing item or mounting facilities and with no appreciable effect on performance or ratings, either electrical or mechanical.

**3.1.14 Mechanical interchangeability.**
The new item's capability of being physically installed and operated in the position previously occupied by the old item without requiring modifications as to mounting holes, etc. Switches, meters, indicators, connectors, etc., should be located as on the previous model, within allowable tolerances. The center-of-gravity of the new item should be the same as in the old item within allowable tolerances.

**3.1.15 Model number.**
A number sequentially assigned for different items requiring the same combination of letters forming a type designation.

**3.1.16 Modification letters.**
A letter assigned in alphabetical sequence starting with the letter “A” to show a modification where interchangeability has been maintained to the extent defined. The letters "I" and "O" will not be used.

**3.1.17 Nomenclature.**
The combination of an approved item name and military type designation, as assigned by the DoDCP.

**3.1.18 Part of:**
a. An item which is required to enable an equipment to fulfill its assigned function should be part of that equipment.



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b. An item which is physically attached to and essential to the operation of another item should be considered part of the item to which it is attached.
c. The item automatically issued, and in all instances with the equipment or item of which it is a part.

**3.1.19 Production assignment.**
A type designation assigned to identify a production equipment, group, or unit. It should consist of the basic type designation assignment modified by deleting the empty parentheses for the initial production item.

**3.1.20 Requesting activity.**
Any activity preparing or forwarding an on-line request for nomenclature (RFN). In ASETDS that is typically the Originator and/or Submitter Review Point.

**3.1.21 Type designation.**
A combination of letters and numerals arranged in a specific sequence to provide a short significant method of identifying the equipment, group, or unit by type.

**3.1.22 Unit.**
In addition to the definition in MIL-HDBK-505, the following applies: A line replaceable unit (LRU) or weapon replaceable assembly (WRA), normally the lowest level of hardware assigned a nomenclature.

**3.1.23 Used with but not part of:**
a. An item which extends the use of an equipment beyond its assigned functions and is to be issued for use with that equipment only under special circumstances is considered as used with but not part of that equipment.
b. An item which may be essential to the operation of another item but is not an integral part thereof, and not permanently attached thereto, is considered as used with but not part of the second item.

**3.1.24 Variable equipment.**
Variable equipment having a variable assemblage that exists under at least one of the following conditions:
a. Those assemblages described as capable of performing more than one function with the functions performed being dependent upon readily exchangeable equipment, groups, units, or combinations thereof, chosen for that installation. Installations may differ by configuration or function, but each installation must be capable of easy and ready conversion to the same function as any other installation. A majority of the items comprising the assemblage, or combinations thereof, must be common to all installations. Each variable configuration should have it's own unique part number or drawing number.
b. Those assemblages which differ between installations due to configuration differences, and may include changes in the quantity or use of items, such as, interconnecting boxes, mounting, or controls. All such assemblages, though physically different, must be functionally and electrically interchangeable.



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c. Those assemblages whose scope of function may be extended or contracted through the addition or deletion of equipment, groups, units, or combinations thereof.

**3.1.25 Variable groups and units.**
A group or unit whose capabilities or functions may be extended or contracted through the addition or deletion of assemblies, subassemblies, parts, or combinations thereof.

**3.2 Acronyms.**
The acronyms used in this standard are defined as follows:

| | |
|---|---|
| ACO | Administrative Contracting Officer |
| ASETDS | Aeronautical and Support Equipment Type Designation System |
| CDRL | Contract Data Requirements List |
| CI | Configuration Item |
| DCP | Departmental Control Point |
| DoDCP | Department of Defense Control Point |
| ECP | Engineering Change Proposal |
| LRU | Line Replaceable Unit |
| PCO | Procurement Contracting Officer |
| PDR | Preliminary Design Review |
| RFN | Request for Nomenclature |
| WRA | Weapon Replaceable Assembly |

## 4. GENERAL REQUIREMENTS

### 4.1 Roles and responsibilities.

**4.1.1 Originator.**
The Originator is the contractor or government representative that starts the RFN process. They initiate the DD Form 61/on-line RFN and ensure it is filled out completely and accurately. Once completed they submit it to the Submitter Review Point. As needed, the originator will work to resolve any deficiences discovered during the subsequent stages of the RFN process.

**4.1.2 Submitter Review Point (SRP).**
The SRP is a government civilian or military representative for the applicable department's project/program office. That individual is knowledgeable about the technical details of the Originator's RFN and reviews the DD Form 61 for accuracy and completeness. The SRP will resolve any deficiencies or errors before submitting the form to the Department Control Point. In some instances a SRP may fulfill both the Originator and SRP roles.

**4.1.3 Department Control Point (DCP).**
The DCP is the official control point within the military department authorized to obtain type designator nomenclature from the Department of Defense Control Point. The DCP will be a government civilian or military representative. The DCP completes an administrate review of the



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DD Form 61 for accuracy and completeness. The DCP will work with the SRP and/or Originator to resolve any deficiencies prior to submission to the Department of Defense Control Point.

**4.1.4 Department of Defenst Control Point (DoDCP).**
The DoDCP is the official assigning agency of type designators for the Department of Defense within the ASETDS system. The DCP will review and research each RFN to ensure correctness and to avoid duplication. The type designation assignment isn't official until signed/approved in ASETDS by the DoDCP. The DoDCP is also the approval authority for any policy waivers pertaining to the content of this publication.

**4.2 Nomenclature assignments.**
The requesting activity shall obtain and use nomenclature in accordance with this standard. Definition of item levels shall be in accordance with this standard and MIL-HDBK-505. The nomenclature of electonic materials are under the guidance of MIL-STD-196 and the Joint Electronics Type Designation Assignment System (JETDAS).

**4.2.1 Digital imaging items.**
A nomenclature shall be required for preproduction and production models of digital imaging equipment, groups, units, accessories, and attachments of photographic items within the scope of 5.1 and subsequent subparagraphs of this standard. The type designations shall be assigned in accordance with 5.1 and subsequent subparagraphs, and shall not apply below the unit level. Accessories and attachments shall be considered units herein.

**4.2.2 Aeronautical equipment, support equipment and munitions.**
A nomenclature shall be required for preproduction and production models of aeronautical and support equipment, units, and groups, within the scope of 5.2 and subsequent subparagraphs of this standard. The type designations shall be assigned in accordance with 5.2 and subsequent subparagraphs and shall not apply below the unit level.

**4.2.3 Aerospace vehicle engine types.**
Engine designations shall be required for preproduction and production models of engine types and motors of all types, within the scope of 5.3 and subsequent subparagraphs of this standard. The engine designations shall be assigned in accordance with 5.3 and subsequent subparagraphs and shall not apply below the complete engine level. Unit level items attached to engines shall be identified in accordance with the procedures in 5.2 and subsequent subparagraphs of this standard, (such as fuel pumps, generators, oil pumps).

**4.3 Nomenclature usage.**
Only nomenclature assigned by the DoDCP shall be used. The appearance of nomenclature for items covered by this standard in invitations for bids, contracts, specifications, drawings and associated documents, shall not constitute official assignment. Prior to the use of such nomenclature (including suffix letters), the requesting activity shall obtain official assignment or confirmation of nomenclature. Assignment of official nomenclature on previous contracts shall not relieve activities of the requirements of this standard. Under no condition shall the manufacturer assume or otherwise receive nomenclature except by official assignment through the procedures outlined herein.



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**4.3.1 Limitation of nomenclature approval.**
The assignment of digital imaging, electronic, or aeronautical and support system nomenclatures shall not constitute approval of any item of equipment, or approval for the use of any particular item in a specific equipment and shall not waive any requirement of the contract involved.

**4.4 USAF nomenclature meeting.**
For U.S. Air Force acquisitions, after the award of contract (normally prior to preliminary design review (PDR), but no later than Critical Design Review (CDR)) a nomenclature meeting should be scheduled with the acquisition activity. The meeting may include representation from the Departmental Control Point for nomenclature as deemed necessary by the acquisition activity. The meeting should discuss the procedures to be followed in the preparation of nomenclature descriptions and the application for and assignment of nomenclature. No nomenclature descriptions shall be prepared prior to the meeting. The acquisition activity shall have the final decision regarding items to be assigned nomenclatures.

**4.5 Application.**
An on-line RFN is system generated for each new configuration item (CI) requiring a type designation, such as equipment, groups, and units. A separate RFN shall be used to request assignment of a new type designation, modification letter, description change, security classification change, or a cancellation. An RFN shall also be submitted upon modification of a CI by an approved engineering change proposal (ECP) which requires a change in type designation. Requests for modification (suffix) letters shall indicate the differences and specify the degree of electrical, mechanical, and functional interchangeability with preceding assignment(s).

**4.5.1 Correctness of request.**
Every effort shall be made by the requesting activity to provide proper recommendations to preclude erroneous assignments. Technical characteristics and functional description data is essential to ensure a timely review of the request by the DCP/DoDCP. Incomplete or incorrect requests for nomenclature shall require clarification from the requesting activity which may result in delays.

**4.6 Notification of assignment.**

**4.6.1 Contractor notification.**
The Government Contracting Officer shall notify the contractor of the official nomenclature assignment. The Government Contracting Officer may be either the Procurement Contracting Officer or the Administrative Contracting Officer.

**4.6.2 DoD activity notification.**
The Departmental Control Point shall notify the DoD activity of the official nomenclature assignment.

**4.7 Coordination.**
Any action requested by a requesting activity against an existing type designated item under design cognizance of another activity shall be coordinated with the original activity for concurrence



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before making the assignment or confirmation. Internal coordination within the respective activities shall be accomplished prior to submission to the DoDCP.

**4.7.1 Modification letters.**
Requests by a department for modification letters to type designations for items which were assigned type designations at the request of another department shall be coordinated by the requesting activity with the original department for concurrence before the modification letter shall be assigned.

**4.8 Cancellation.**
The originating department shall cancel nomenclature and designations upon request when the following conditions exist:

a. There has been no procurement of the item.
b. No experimental models exist.
c. No further use of the type designation is required for developmental purposes.
d. The item is no longer in a service inventory.
e. When an existing assignment is changed to a variable configuration or vise versa.

**4.8.1 Reissue.**
A nomenclature or designation cancelled in accordance with 4.8 of this standard shall not be reissued to identify another item.

**4.8.2 Reactivation of nomenclature and designations.**
Cancelled nomenclature and designations shall not be reactivated except upon request of, or coordinated approval of, the department that originally requested the nomenclature or designation.

**4.9 Justification of designation.**
Each type designation shall be justified on the basis of a description, which shall contain electrical, mechanical and reference data to distinguish the item described from all other items. The same type designation shall not be used to identify items which differ in operational, electrical, or mechanical characteristics.

**4.9.1 Description.**
All requests for the assignment of type designations shall contain a description of the item (see 6.3). The descriptive data shall consist of the technical and functional description and other informational data required under the heading Technical Data (blocks 14 thru 24) on the on-line RFN. The descriptive data shall convey condensed, factual data relating to the principle descriptive and identifying characteristics and features of the item concerned. The data required shall be informational rather than instructional, and to the extent and detail specified herein, shall be complete without reference to other publications for essential details.

**4.9.1.1 Technical description.**
The technical description shall consist of the technical and specification characteristics and features that directly or indirectly convey and contribute to an accurate interpretation and recognition of the item submitted. The technical data submitted shall support the item name



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selected for the item. The type designation shall be determined by the technical characteristics of the item and not by its assigned item name.

**4.9.1.2 Functional description.**
The functional description shall be presented in narrative form and shall support the recommended type designation.

**4.10 Changes to type designation.**
Whenever an engineering change or other modification, which affects a substitute item or interchangeable item status, as defined in MIL-HDBK-505, is to be implemented in a type designated preproduction or production model, the type designation shall be changed and the requesting activity shall obtain and use the changed type designation (see 4.10.1 through 4.10.5).

**4.10.1 Type designation change when a modified unit is not interchangeable with a previous unit.**
When the modified unit is not an interchangeable item with the previous unit, a new type designation shall be assigned.

**4.10.2 Type designation change when a modified unit becomes a substitute item for all previous units.**
When the modified unit becomes a substitute item for the previous unit, a suffix letter shall be added to the type designation. For example, a KA-63A would become KA-63B; an RT-635/APN would become RT-635A/APN; and an AAU-34/A would become AAU-34A/A.

**4.10.3 Type designation change to higher level elements when modified.**
Whenever a type designated unit is modified in accordance with 4.10.1 or 4.10.2, the type designation of its next higher element and all progressively higher elements shall be changed, up to and including the element where interchangeability or substitutability is maintained. If interchangeability or substitutability of the higher element is maintained, the type designation shall be changed by adding a suffix letter to indicate that a modification has been made below the higher element.

**4.10.4 Nomenclature assignment requests for previously nomenclatured items which are modified.**
Modifications to units which have been assigned nomenclature shall require the submittal of an on-line RFN for each unit modified, including an on-line RFN for the group and equipment, as applicable, of which the unit is a part. The above applies only to modifications which affect substitute or interchangeable item status as defined in MIL-HDBK-505.

**4.10.5 Nomenclature requests for items modified after delivery.**
On-line RFNs shall be submitted by the requesting activity for items which are modified as described in 4.10, provided such items require a nomenclature in accordance with this standard.

**4.11 Consecutive numbering.**
a. The skipping of numbers in type designation assignments shall not be permitted.



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b. Blocks of number reservations shall not be permitted without prior justification and approval.

**4.12 On-line request for nomenclature (RFN) security.**
ASETDS is not a secure database and therefore will contain no classified information. However, nomenclatures can and will be assigned to classified items. It is the responsibility of the requesting activity to ensure that no CLASSIFIED information is submitted on the RFN.

**4.12.1 Unclassified requests.**
Unclassified requests for nomenclature shall be submitted whenever possible. Data contained in the on-line RFN shall be tailored to accurately describe the item. However, a conscientious effort shall be made by the requestor to preclude inclusion of classified data on the requests. Departmental Control Points shall establish procedures in coordination with the DoD Control Point to process requests which contain limited technical descriptions in order to preclude processing of classified requests.

**4.12.2 Security.**
All classified requests for nomenclature shall be prepared and maintained in accordance with DoDI 5220.22, or DoDM 5200.01, as applicable.

**4.12.3 Nomenclature security.**
The item name, type designation, and combination of, shall be UNCLASSIFIED.

**4.12.4 Regrading existing technical data.**
Regrading of existing classified type designation technical data shall be accomplished through the submission of a new on-line RFN by the requesting activity. All regrading submissions shall include the appropriate downgrading and declassification markings required by the documents specified in 4.12.2.

**4.13 Retroactive application.**
The type designation systems covered by this standard shall not be retroactively applied to items already type designated under other systems. On-line RFNs shall not be submitted retroactively for the assignment of an approved nomenclature to equipment already in the supply inventory and where a federal stock number has already been assigned. The previously assigned nomenclatures shall be used until the physical items have been attrited from the supply system. This shall not preclude the assignment of nomenclatures to revised or modified equipment.

**4.14 One identification.**
One individual type designation shall be assigned to each item covered by this standard. Once a type designation has been assigned, it shall not be used to identify any other item, even if the original assignment is cancelled.

**4.15 Type designation usage and abbreviations.**
Type designations assigned by the DoDCP shall be used strictly as assigned. At the discretion of the responsible department, abbreviations conforming to ASME Y14.38 shall be used where space



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is not sufficient for spelling out an item name. However, the basic noun or noun phrase shall be spelled out.

**4.16 Request for basic assignment.**
Requests for basic assignments may be submitted at any time, but preferably at the time that a research and development project reaches the end of the study and investigation phase of the technical program.

**4.16.1 Basic assignment.**
A type designation shall be assigned to identify the general design of the equipment, component, or unit. It shall consist of a type designation with ***an empty parentheses directly*** following the model number. ***Examples: A/S32P-5( ), SVG-7( )/A24G-6, or SUU-9( )/A***. A basic assignment should be cancelled when/if it progresses to a development or production assigment.

**4.16.2 Development assignment.**
A type designation shall be assigned by the development activity to identify a development equipment, group, or unit after first obtaining a basic assignment from the DoDCP. It shall consist of the basic assignment and the appropriate development organization-indicator (see Appendix A) followed by a numeral placed within the empty parentheses of a basic assignment. These numerals shall be assigned in sequence starting with "-1" for each basic assignment. Examples of development assignments are: AN/ARC-123(XA-1), A/S32P-5(XA-1); R-765(XA-I)/A, SUU-9(XA-I)/A for the first development model and AN/ARC-123(XA-2), A/S32P-5(XA-2) etc., for the second development model. The assigning and recording of specific numerals after an experimental indicator, within the parentheses of a type designation, shall be the development activity's responsibility without preference to the DoD Control Point. A development assignment should be cancelled when/if it progresses to a production assignment.

**4.16.3 Production assignment.**
A type designation shall be assigned to identify a production equipment, group, or unit. It shall consist of the basic type designation modified by deleting the empty parentheses for the initial production item. ***Examples: A/S32P-5, RT-1234/ARC-123, etc***.

**4.17 Revisions.**
When the description of the item is no longer technically correct, the requesting activity shall submit a revised request containing the data that accurately reflects the item being procured. In addition, the revised request shall include a statement, in block 24 of the on-line RFN, that no items were produced and delivered to the Government as described under the basic or preceding request. If otherwise, the request shall indicate the differences in the models and shall specify the degree of interchangeability or substitutability of each model as compared with the basic, and every preceding model. This request shall indicate the serial number of the first item affected by the change.

**4.18 Reservations.**
The DoDCP can grant reservations for type designations on a case-by case basis (see 4.11). The need for type designations shall be an emergency or high priority requirement, as determined by



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the requesting activities, for which information is available "or item name and type designation recommendation".

## 5. DETAILED REQUIREMENTS

### 5.1 Digital imaging equipment.

This section covers type designation of conventional photographic equipment. It does not include electro-optical or other imaging techniques which shall be in accordance with MIL-STD-196.

#### 5.1.1 Digital imaging nomenclature.

The type designations assigned under this system shall be used with the approved item name to form the complete nomenclature. In each instance, the item name shall precede the type designation.

#### 5.1.2 Application of digital imaging type designations.

Photographic type designations shall be assigned to the following:

a. Complete items, groups, sets or systems of photographic equipment designed for military use.
b. Components, accessories, and attachments for photographic equipment designed for military use.
c. Commercial photographic articles adopted for military use.

#### 5.1.3 Categories.

For the purpose of this type designation system, photographic equipment shall be divided into three categories, as follows:

a. Digital camera.
b. Digital image-processing software.
c. Picture-using equipment.

#### 5.1.4 Type designation.

The type designations shall consist of five parts and a dash. These shall be the two letter code, the dash, the model number, the suffix letter, and when required, the production issue number. Figure 1 shows a graphical illustration breakdown of a designation.

**FIGURE 1. Breakdown of photographic type designations.**

| KA- | 7 | ( ) | A | 1 |
|---|---|---|---|---|
| Two letter code | Model number | Basic assignment[1/] | Suffix letter | Production issue number |

**Two letter code.**

Each category is assigned a letter designation, indicating major item, accessory, attachment, or component. Likewise, the general mission or function of each item shall be assigned a letter designation. The category letter and the mission letter together shall form the two letter code portion of the type designation (see Appendix B).



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**Model numbers.**
Each two-letter combination shall have a separate series of model numbers assigned in sequence, beginning at "1", i.e. KA-1, KA-2.

**Basic assignment.**
Following the model number for each basic assignment, there shall be an empty parentheses, denoting the assignment is a basic assignment, The basic assignment shall not identify a specific equipment, but shall identify all equipment of the same general design.

**Suffix letters.**
The suffix letter "A" is assigned to the type designation, in lieu of the empty parentheses, to indicate the first production assignment. Suffix letters shall be assigned in alphabetical sequence.

**Production issue numbers.**
Production issue numbers, when required, are assigned to a given model number and suffix letter when a manufacturer uses different maintenance parts. Interchangeability is required for an issue number. All versions of a given suffix letter must be interchangeable.

**Examples of type designation sequence:**

| | |
|---|---|
| KA-1( ) | Basic assignment. |
| KA-1A | First production assignment. |
| KA-1A1 | Second production assignment (Different maintenance parts only from KA-1A). |
| KA-1A2 | Third production assignment (Different manufacturer, utilizing different maintenance parts lists for either KA-1A or KA-1A1). |
| KA-1B | Subsequent production assignment. (Manufacturer modification of KA-1A, including one way progressive substitutability with the A-1A). |
| KA-1B1 | Subsequent production assignment. (Different maintenance parts for KA-1B). |

1/ Only used on experimental or developmental assignments when complete technical data is not available.

**FIGURE 1. Breakdown of photographic type designations - Continued.**

### 5.1.5 Modifications.
Modification or improvements which affect electrical, physical, or functional interchangeability shall require the next number available for that particular two letter combination, provided the mission shall not be changed; otherwise an entirely new type designation shall be assigned under the applicable two letter combination.

### 5.1.6 Submission of requests for nomenclature/type designation actions.

#### 5.1.6.1 Submission of requests for nomenclature/type designation actions.
Requests for nomenclature/type designation actions to digital imaging equipment shall be submitted electronically in ASETDS at the following web site: https://tdas7.apg.army.mil/ASETDS//. Submissions will be routed to the applicable Submitter



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Review Point, Department Control Point and the DoD Control Point. Proper selections in block 2b of the on-line RFN will help ensure prompt processing of requests.

### 5.1.7 Centralized administration.

#### 5.1.7.1 DoD Control Point.

Aeronautical and Support Equipment, AFLCMC/EZSC, Bldg. 28, Wright-Patterson AFB, OH 45433-7017 shall be designated as the DoD Control Point for items type designated under the Aeronautic and Support Equipment Type Designation System (ASETDS).

## 5.2 Aeronautical equipment, support equipment, and munitions.

### 5.2.1 Aeronautical equipment, support equipment, and munitions nomenclature.

The type designations assigned under this system shall be used with the item name to form the complete nomenclature. In each instance, the item name shall precede the type designation.

### 5.2.2 Application of aeronautical equipment, support equipment, and munitions type designations.

Aeronautical equipment, support equipment, and munitions type designations shall be assigned to the following:

a. Complete equipment, groups, and units designed for the military departments.
b. Commercial equipment, groups, and units where military identification shall be required for use by Government activities.

### 5.2.3 Categories within the ASETDS.

Aeronautical equipment,support, equipment and munitions covered by this system shall be divided into three categories: equipment, groups, and units; with the manner of assigning type designations differing somewhat for each category.

### 5.2.4 Equipment type designations.

These designations shall consist of an “A/” to denote assignment in the ASETDS; an installation letter, a two-digit numerical type-of-equipment indicator, a purpose letter followed by a dash (-), a numerical model number, and modification letter when applicable. Figure 2 shows a graphical illustration breakdown of an equipment designation.



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| A/ | S | 32 | P_ | 5 | ( ) | A |
|---|---|---|---|---|---|---|
| Aero/support system | Installation letter | Type indicator | Purpose letter | Model number | Basic assignment[1/] | Suffix letter |

**Installation letter.**
Each installation shall be assigned a distinctive letter designation to indicate where the equipment shall be installed (see Appendix C, Table C-I, column 1).

**Type indicator.**
Each type-of-equipment indicator shall be assigned a distinctive two-digit designation to indicate the type-of-equipment (see Appendix C, Table C-I, column 2).

**Purpose letter.**
Each purpose shall be assigned a distinctive letter designation to indicate the purpose of the equipment (see Appendix C, Table C-I, column 3).

**Model number.**
Each installation, type-of-equipment, and purpose letter combination shall have a separate series of model numbers assigned in sequence, beginning at "1" (such as A/S32P-1).

**Suffix letters.**
A suffix letter shall be assigned when the equipment is modified in a manner that interchangeability is maintained. The suffix letter shall be assigned in alphabetical sequence (such as A/S32P-5A or A/A24B-5B).

**Basic assignment.**
Following the model number for each basic assignment shall be an empty parentheses, denoting the assignment is a basic assignment. The basic assignment shall not identify a specific equipment, but shall identify all equipment of the same general design.

1/ Only used an experimental or developmental assignments when complete technical data is not available.

**FIGURE 2. Breakdown of A/ equipment designation.**

### 5.2.5 Designation changes.

Changes to equipment designations shall occur only as follows:

a. When a complete equipment is modified or improved and it maintains functional, electrical, and physical interchangeability, the equipment shall be identified by the assignment of a new suffix letter.
b. When a complete equipment is modified or improved to the extent that either its electrical, physical, or functional interchangeability is affected, the equipment shall be assigned the next model number available for that particular installation-type-purpose combination.
c. When the installation of AN equipment, the type of AN equipment, or the purpose of AN equipment is changed, a new type designation shall be assigned.

### 5.2.6 Group type designation.

All groups shall be identified by a two-letter code selected from Appendix D, followed by a "G", a dash, a model number, a slant bar, and an equipment designator. Applicable equipment indicator letters following the slant bar shall be selected from Appendix C after considering the potential of the group for multiple or peculiar application (such as SVG-7/A, SVG-7/A24, SVG-7/A24A, or SVG-7/A24A-3).

### 5.2.7 Unit type designations.

Units shall be designated by a two-letter code taken from Appendix D, followed by the letter "U", a dash, a model number, a slant bar, and an equipment designator. Applicable equipment indicator letters following the slant bar shall be selected from Appendix C after considering the potential of the unit for multiple or peculiar application (i.e., SVU-5/A, SVU-5/A24, SVU-5/A24A, or SVU-5/A24A-3).



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**5.2.8 Changes to group or unit designation.**
A group or unit shall be physically, electrically, and functionally interchangeable in order to maintain its original identity.

**5.2.8.1 Group interchangeability.**
When subassemblies, and units, are no longer interchangeable, but the group itself remains physically, electrically, and functionally interchangeable with the unmodified item, a suffix letter shall be assigned.

**5.2.8.2 Group or unit noninterchangeability.**
When a group or unit is modified or improved to the extent that either its electrical, physical, or functional interchangeability is affected, then the group or unit shall be assigned the next model number available for that particular two-letter code.

**5.2.8.3 Groups or units that are part of two or more equipments.**
Groups or units that are part of two or more equipments shall be identified in the usual manner, except that after the slant bar, there shall appear the appropriate installation-type-purpose indicator codes without an equipment model number. Thus, a servo-mechanism group or unit that is part of the A/A24G-5, A/A24G-7, and A/A24G-15 would be identified as SVG-16/A24G or SVU-8/A24G.

**5.2.9 Equipment designed for training.**
Equipment designed for training purposes shall be assigned type designations dependent upon their relationship to functional equipment.

**5.2.9.1 Equipment to provide training on a specific basic equipment.**
Equipment designed to provide training on specific equipment shall be assigned the specific equipment type designation, followed by a dash, the letter "T", and a number. Example: Automatic Pilot Trainer A/A24G-6-T1 would be the first training equipment for Automatic Pilot A/A24G-6.

**5.2.9.2 Equipment to provide training on general equipment.**
Equipment designed to provide training on general types of equipment shall be assigned the general equipment indicators followed by a dash, the letter "T", and a number. Example: Automatic Pilot Trainer A/A24G-T1 would be the first equipment for general automatic pilot training.

**5.2.9.3 Miscellaneous training equipment.**
Complete equipment designed to provide training on types of equipment not covered by this system, (such as aircraft engines) shall be designated as follows: A/ applicable installation letter, type indicators, and the purpose letter "U" followed by a -T1, -T2, -T3, etc., (see Appendix C). Example: Trainer, Aircraft Engine A/E32U-T1.

**5.2.10 Units and groups designed for training.**
Training versions of units and groups shall be identified by the letter "T" followed by a version dash number, enclosed in parentheses and placed between the type number and the slant bar. For example, the first training version of the ACU-99/A34 would be identified as ACU-99(T-1)/A34,



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the second version ACU-99(T-2)/A34. Similarly, the third training version of the GEG-75/E24A would be identified as GEG-75(T-3)/E24A.

**5.2.11 Variable type designation.**
Complete equipment, groups, and units with variable component lists shall be assigned type designations in the same manner as for definitive equipment, components, and units, except the letter "V", enclosed in parentheses, shall be added to the type designation immediately following the model number (e.g., A/F42P-5(V), RYG-59(V)/U, SDU-35(V)/P).

**5.2.12 Specific configurations of variable items.**
Variable equipment, groups, and units shall have a number assigned following the parenthetical V (V) when further identification is required to identify specific configurations of a variable item, such as A/F42P-5(V)1, A/F42P-5(V)2, RYG-59(V)1/U, RYG-59(V)2/U, SDU-35(V)1/P, SDU-35(V)2/P. The complement data listing for a specific configuration shows the precise quantity of equipment selected from the complement listing of the basic (V). A separate request for nomenclature is required for each specific configuration requested.

**5.2.13 Munitions items.**
The following exceptions shall apply to the assignment of type designations to munitions items.

**5.2.13.1 Non-nuclear munitions designations.**
All airdropped non-nuclear items shall be type designated as units in accordance with 5.2.7. Equipment designations and group designations shall not be applied to airdropped non-nuclear munitions items.

**5.2.13.2 Type and purpose indicator.**
Only the installation letter (see Appendix C, column 1), shall be utilized following the slant bar. The type and purpose indicators shall not be applied to airdropped non-nuclear munitions items, such as BLU-97/B.

**5.2.14 Dummy and training versions of live munitions.**
Test, practice, and training versions of live munitions shall be categorized as either dummy or training.

**5.2.14.1 Dummy versions of live munitions shall be completely inert.**
Dummy munitions shall be used for display purposes, testing, and training operations (assembly, loading, handling, and dry-run operations).

**5.2.14.2 Training versions of live munitions shall not be inert.**
Munitions classified as training include some type of explosive, burning, or smoke producing element for ground impact marking or other training purposes in association with firing, flying, prepositioning, and dropping operations.



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**5.2.14.3 Identification of dummy versions of live munitions.**
Dummy versions of live munitions shall be identified in a similar manner to training versions, except the letter "D" shall be used in place of the letter "T" and the number following the letter "D" shall have a single, specific meaning as follows:

a. **D-1 Parent inert:**
An inert version of the operational munition.

b. **D-2 Load Crew:**
Of the same diameter and shape as the parent inert. It shall have a capability of duplicating every action required during the loading of the parent munition. Internal hardware shall not be necessary, except as may be required for checkout. The weight shall be within 10 percent and the center-of-gravity shall be within 2 percent of the parent munition.

c. **D-3 Explosive ordnance disposal (EOD):**
Identical to the inert parent munitions except that easily replaceable parts shall be built into or on the munition where EOD procedures indicate destructive type of work shall be needed in EOD training.

d. **D-4 Ballasted:**
An inert dispenser or end item munition body, ballasted to 90 to 100 percent of weight of the parent munition, and true to shape and center-of-gravity (within 2 percent), of the parent munition. External and internal aircraft/munition interface and checkout during loading or handling and shall utilize parent munition technical order for loading. It shall be compatible with applicable service loading and handling equipment.

e. **D-5 Miscellaneous:**
Any dummy application not covered by one of the four standard categories.

**5.2.14.4 Identification of training versions of live munitions.**
Training versions of live munitions shall be identified by the letter "T" followed by a version dash number, enclosed in parentheses, and placed between the type number and the slant bar. A separate sequence of version numbers shall be used for training munitions, each starting at -1 for each type designated item. For example, the first training version of the BLU-99/B would be identified as BLU-99(T-1)/B, the second version BLU-99(T-2)/B. Similarly, the third training version of the BLU-75/B would be identified as BLU-75(T-3)/B.

**5.2.15 Dummy and training versions of munitions with no live counterparts.**
Training and dummy versions of munitions for which there is no live munition (parent), or when any association between the actual live munition with its test and training shape is classified, shall be identified by the appropriate "T" or "D" identifier in the same manner enumerated in 5.2.14.1 or 5.2.14.2.

**5.2.16 Submission of requests for nomenclature/type designation action.**

**5.2.16.1 Submission of requests for nomenclature/type designation actions**
Requests for nomenclature/type designation actions to aeronautical equipment, support equipment and munitions shall be submitted electronically in ASETDS at the following web site: https://tdas7.apg.army.mil/ASETDS//. Submissions will be routed to the applicable Submitter



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Review Point, Department Control Point and the DoD Control Point. Proper selections in block 2b of the on-line RFN will help ensure prompt processing of requests.

**5.2.17 Centralized administration.**

**5.2.17.1 DoD control point.**
Aeronautical and Support Equipment, AFLCMC/EZSC, Bldg. 28, Wright-Patterson AFB, OH 45433-7017, shall be designated as the DoDCP for items type designated under this system.

**5.3 Aerospace vehicle engine types.**

**5.3.1 New designation.**
A new engine designation shall be assigned when, in the judgement of the assigning service, any one of the following conditions shall be met:

a. A need (either administrative or technical), exists for reference to, or definition of, a particular engine configuration or design which has not previously been assigned a designation. Normally, this shall occur as a result of new engine development programs, or upon the initial entry into the services inventory of off-the-shelf engines, and shall involve a newly assigned type indicator.
b. An engine configuration or design with an existing designation shall be so revised that interchangeability of the complete engine in an aircraft shall be affected. Or, there shall be a significant change in performance or installation, and the change shall not be made immediately retroactive to all engines of the same designation previously delivered. Ordinarily, this shall involve the assignment of a new model number only. However, when the design change alters the engine fundamentally, a new type indicator may be assigned.
c. Design changes shall be made to an engine, with an existing designation, to the extent that a substantial number of parts or subassemblies (or both) shall be no longer interchangeable with the corresponding parts or subassemblies furnished in accordance with the approved parts list for the original model engine. But the changes shall not affect the performance, installation, or interchangeability of the complete engine in an aircraft. Ordinarily, this shall involve only the addition or change of the suffix letter in the model indicator, although change of the model number may be indicated if the quantity of parts affected is sufficiently large.

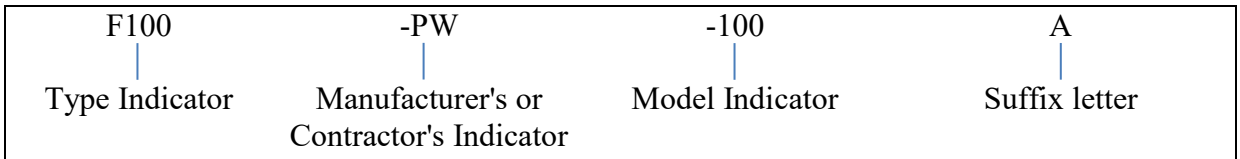
**5.3.2 Application of aerospace vehicle propulsion engine type designations.**
The procedures set forth in this standard shall be applicable to developmental, preproduction, and production propulsion devices. The type designation shall always apply to one specific propulsion device or subsequent models thereof, as indicated by the model number.

**5.3.3 Assigned designation.**
A specific propulsion device configuration or design shall have only one designation. Should one department desire to use a propulsion device developed by another department without change, the original designation assigned shall be used. In the case where production changes are required,



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the type indicator and manufacturer's indicator shall remain unchanged. The model indicator shall be in accordance with Figure 3.




**Type indicator.**
The type indicator part of the engine designation shall consist of the appropriate type letter symbol selected from Appendix F, Table F-I, together with a type numeral. The type numerals used in connection with the type letter shall be assigned consecutively by the services, and shall begin with the number 100 for the Air Force, the number 400 for the Navy, and the number 700 for the Army. The type numbers shall be arbitrary, and shall not represent any characteristics of the units involved.

**Manufacturer's and contractor's indicator.**
The manufacturer's or contractor's indicator shall consist of a dash and a manufacturer's code selected from Appendix F, Table F-III.

**Model indicator.**
The model indicator shall consist of a dash and a model number. The model number shall be assigned for a specific configuration of a given type engine. Air Force model numbers for each type of engine shall begin with -100 and shall continue with consecutive numbers. Navy model numbers for each type of engine shall begin with -400 and shall continue with consecutive numbers. Army model numbers for each type of engine shall begin with -700 and shall continue with consecutive numbers.

**Suffix letter changes to designated engines.**
When an engine with an existing designation requires the addition of a suffix letter as a result of minor design changes, the engine shall continue to use the type indicator and model number initially assigned. The change shall be reflected by the addition of the suffix letter "A" or the next consecutive suffix letter for the specific engine, except for the letters "I", "O", and "W". (Example: When modified, the R100-PW-100 would become the R100-PW-100A, the F110-GE-100A would become the F110-GE-100B.)

**NOTES:**
1. The letters "I" and "O" shall not be used.
2. The letter "W" shall be used with the basic model numeral, or numeral and suffix letter, to indicate engines equipped with water injection. When so used, the letter "W" shall precede any other suffix letter assigned in the model indicator.

**FIGURE 3. Breakdown of an air-breathing engine designation.**

### 5.3.4 Numeric designations.
A new numerical designation shall be assigned when an engine design is new or so revised that a significant change in performance, flight safety, installation, or interchangeability of the complete engine shall be affected, and the change shall not be made retroactive to all engines of the same model previously delivered.



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**5.3.5 Former engine designation assignments.**
An engine initially designated in accordance with the former Air Force-Navy aeronautical system (now cancelled), shall retain the designation assigned for the remainder of its in-service life. Designation changes to engines so designated shall be permitted only to the extent specified in 5.3.6 or 5.3.10.

**5.3.6 Mixed designation.**
If a new engine model number is required , the new model number shall be assigned as described in this standard or, at the option of the issuing department, the initial model number in the new series shall consist of the last model number added to the base starting number of the range assigned to the issuing department. Thereafter, subsequent model designations shall be assigned in consecutive order.

**5.3.7 Special designation.**
The letters "X" and "Y" may be used at the discretion of the services for the purpose of signifying development engines, as applicable. Generally, the prefix letter "X" (experimental) shall be used to denote an engine under development which has not reached the stage of successfully completing the official preliminary flight rating tests. The prefix "Y" (prototype) shall be used to denote an engine still in the development category, but which has been cleared for limited flight use by successfully completing the preliminary flight rating tests. When used, such letters shall precede the designation arrangement described herein.

**5.3.8 Manufacturer's and contractor's code symbol.**
The manufacturer's and contractor's code symbol shall be a two-letter designation which identifies the manufacturer or contractor (see Appendix E). These two-letter symbols shall not be confused with those codes established by the Federal Supply Code for Manufacturers, H4, which covers code identification symbols of principal manufacturers of equipment and parts.

**5.3.9 Special manufacturers' or contractors' symbols.**
In special cases, such as where two manufacturers or contractors are operating in a joint effort, a special manufacturers' or contractors' symbol may be assigned which combines a letter of each of the two individual manufacturer's or contractor's symbols. Engines produced by one of the two manufacturers or contractors shall use only that manufacturer's or contractor's symbol.

**5.3.10 Suffix letters.**
Suffix letters shall be assigned consecutively, starting with the letter “A”. The letters “I” and “O” shall be skipped to preclude confusion with the numerals 1 (one) and 0 (zero). The letter “W” is reserved for use on engines equipped with water injection.

**5.3.11 Major categories.**
This standard covers the designation of air-breathing type aerospace propulsion devices. Non-air breathing devices, to include rocket mortars, rocket engines, hybrids or other propulsion devices which do not utilize atmospheric oxygen for combustion, are designated under the provisions of AFI 16-401, Designating and Naming Defense Military Aerospace Vehicles.



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The air-breathing engine designation categories shall include adaptive, reciprocating, turbo (turbojet, turbofan, turboprop, and turboshaft), and any other devices which utilize atmospheric oxygen for combustion.

**5.3.11.1 Air-breathing engine designations.**
Air-breathing engine designators shall consist of a type indicator, manufacturer's indicator, and the model indicator. Prefix and suffix letters shall be added as necessary (see 5.3.7 and 5.3.10). Figure 3 shows a graphical illustration breakdown of an air-breathing engine designation.

**5.3.11.1.1 Assignment responsibility.**
Each service shall be responsible for the assignment of the air-breathing engine designations within the scope of this standard. For Air Force submissions, request shall be submitted to the DCP via a signed memorandum that includes the technical details and characteristics of the engine. When a designation is assigned, the assigning service shall notify the other two services of this action and include a definitive description of the engine so designated. Notification letters shall be forwarded as follows:

Commander
US Army Communications-Electronics Command
ATTN: AMSEL-LCL-SII
Bldg 6565 Surveillance Loop
Aberdeen Proving Ground, MD 21005-1845

Commanding Officer
Naval Air Systems Command
ATTN: AIR-4.4.1.2
Patuxent River, MD 20670

AFLCMC/EZSC
2145 Monahan Way, Bldg. 28
Wright-Patterson AFB, OH 45433-7017

**5.3.11.1.2 Use of assigned-type designation.**
The authorized type designation shall be used strictly as assigned. An assignment may be changed upon the request of the initiating activity, provided that such a change shall not be contrary to established policy. Where necessary, item names may be omitted from identification markings on equipment at the discretion of the responsible department.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

### 6.1 Intended use.

This standard establishes the requirements for obtaining official assignment of nomenclature and conformation of existing nomenclature for items of photographic, aeronautical equipment, and support equipment acquired by the DoD.



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### 6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of this standard.

### 6.3 Associated Data Item Descriptions (DIDs).

This standard has been assigned an Acqusition Management Systems Control (AMSC) number authorizing it as the source document for the following DIDs. When it is necessary to obtain the data, the applicable DIDs must be listed on the Contract Data Requirements List (DD Form 1423).

| DID Number | DID Title |
|---|---|
| DI-SESS-81254 | Request for Nomenclature (DD Form 61) |

The above DIDs were current as of the date of this standard. The ASSIST database should be researched at https://quicksearch.dla.mil to ensure that only current and approved DIDs are cited on the DD Form 1423.

### 6.4 Subject term (key word) listing.

Air breathing engines
Item name
Munitions
Nomenclature
Ordnance

### 6.5 International interest.

Paragraphs 6.6 through 6.6.1.9 of this standard have been agreed to by the U.S. Departments of Defense and the Canadian Department of National Defence. ***The acceptance of these paragraphs are contained in OASD(AR) Memorandum for ASD(AFSC) dated 12 October 1969***, Subject: Canadian Forces Participation in Type Designation Systems for Equipment and Supplies - Aircraft and Photographic. When change notice, revision, or cancellation of this standard is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

### 6.6 Canadian integration.

#### 6.6.1 Canadian nomenclature assignments.

Canadian requests for type designations will be assigned and registered by the Canadian Forces Headquarters (CFH), Department of National Defence, Canada, in conformance with the policies contained in this standard.



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**6.6.1.1 Notification.**
The CFH will notify the assigning activity of all assignments and that body will confirm such assignments. Where an American assignment has previously been made, the Canadian assignment will be cancelled in favor of the previous assignment.

**6.6.1.2 Distribution.**
The CFH will transmit to the assigning activity, copies of the descriptive details, of each Canadian unclassified type designation assignment on the Canadian equivalent of the RFN.

**6.6.1.3 U.S. modifications to Canadian items.**
Requests by United States activities for modification letter assignments to Canadian equipment will be coordinated with the CFH and will be assigned from the Canadian register.

**6.6.1.4 Canadian modifications to U.S. items.**
Requests by Canadian agencies for modification letter assignments to United States equipment will be coordinated by the assigning activity with the cognizant service and will be assigned from the United States register.

**6.6.1.5 Distribution of technical data.**
The CFH will be on the official distribution list for unclassified technical data.

**6.6.1.6 Confidential and secret items.**
Type designation assignments for confidential and secret items will be made known, but classified descriptive details will be passed only upon approval of requests on an individual basis.

**6.6.1.7 Copies of technical data.**
The CFH will be furnished the required number of copies of unclassified technical data for distribution within the Canadian Department of National Defence.

**6.6.1.8 Equipment numbers.**
The block of model numbers from 500 to 599 and 2500 to 2599 will be used by the CFH in the assignment of equipment nomenclature.

**6.6.1.9 Group and unit numbers.**
The block of group and unit numbers from 5,000 to 5,999 and 25,000 to 25,999 will be used by the CFH in the assignment of group and unit type designations.

**6.7 Supersession data.**
The following documents are superseded:

| | |
|---|---|
| MIL-STD-155 | Joint Photographic Type Designation System |
| MIL-STD-815 | Type Designation System for Rocket Engines, Motors, and Hybrids |
| MIL-STD-842(USAF) | Uniform Procedures for Type Designation and Item Identification |



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| | |
|---|---|
| MIL-STD-875 | Type Designation System for Aeronautical and Support Equipment |
| MIL-STD-879 | Designation of Aircraft Propulsion Gas Turbine Engines |
| MIL-STD-1557 | Designation of Aircraft Propulsion Reciprocating Engines |
| MIL-N-7513(USAF) | Nomenclature Assignment, Contractor's Method of Obtaining |

**6.8 Changes from previous issue.**
Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.



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# APPENDIX A

## DEVELOPMENT INDICATORS FOR IDENTIFICATION OF DEVELOPMENT ACTIVITIES

### A.1 SCOPE

**A.1.1 Scope.**
This appendix lists the development indicators which are to be used to identify a development equipment, group, or unit for the preparation of nomenclature or type designation requests.

### A.2 APPLICABLE DOCUMENTS
This section does not apply.

### A.3 DEFINITIONS AND ACRONYMS
This section does not apply.

### A.4 GENERAL REQUIREMENTS

**A.4.1 List of tables.**
The following tables list the development indicators which are to be used to identify a development equipment, group, or unit (see 4.16.2).

a. Table A- I Air Force development indicators for developing activities.
b. Table A- II Army development indicators for developing activities.
c. Table A- III Navy development indicators for developing activities.
d. Table A- IV Other development indicators for developing activities.



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**TABLE A- I. Air Force development indicators for developing activities.**

| AIR FORCE | |
|---|---|
| XA | Aeronautical and Support Equipment, AFLCMC/EZSC, Wright-Patterson AFB, OH 45433-7017 |
| XD | Electronics Systems Division, Hanscom AFB, MA 01731 |
| XI | Munitions Systems Division, Eglin AFB, Fl, 32542 |
| XV | Air Force Weapons Laboratory, Kirtland AFB, NM 87117 |
| XW | Rome Air Development Center, Griffiss AFB, NY 13441 |
| XBS | Ballistic Systems Division, Norton AFB, CA 92409 |
| XSS | Space Systems Division, Los Angeles AFB, CA 90009-2960 |
| The following Air Force designators are to be used for assigning additional designations to previously type designated items. For all new equipment use XA. | |
| XH | Aerial Reconnaissance Laboratory, Wright-Patterson AFB, OH 4S433 |
| XK | Flight Control Laboratory, Wright-Patterson AFB, OH 45433 |
| XQ | Aeronautical Accessories Laboratory, Wright-Patterson AFB, OH 45433 |
| XS | Electronic Component Laboratory, Wright-Patterson AFB, OH 45433 |
| XY | Weapons Guidance Laboratory, Wright-Patterson AFB, OH 45433 |
| Inactive | |
| XCC | Air Force Missile Test Center, Patrick AFB, FL 32925 |

**TABLE A- II. Army development indicators for developing activities.**

| ARMY | |
|---|---|
| XC | U.S. Army Communications-Electronics Command (CECOM), Fort Monmouth, NJ 07703-5006 |
| XO | U.S. Army, Missile Command (MICOM) Redstone Arsenal, AL 35898-5000 |
| XT | U.S. Army Intelligence & Security Command (INSCOM) Arlington Hall Station, VA 22212 |
| XAV | U.S. Army Aviation Research & Development Command (AVRADA), Fort Monmouth, NJ 07703 |
| XER | U.S. Army Electronics Research & Development Command, (ERADCOM) Adelphi, MD 20783 |
| XME | U.S. Army Mobility Equipment Research & Development Command, (MERADCOM) Fort Belvoir, VA 22060 |
| Inactive | |
| XE | U.S. Army Electronics Laboratory, Fort Monmouth, NJ 07703 |
| XF | Frankford Arsenal, Philadelphia, PA 19104 |
| XL | U.S. Army Signal Electronics Research Unit, Mountain View, CA 94042 |
| XM | U.S. Army Signal Engineering Laboratories, Hexagon, Fort Monmouth, NJ 07763 |
| XAE | U.S. Army Research & Development Activity, Fort Huachuca, AZ 85613 |



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**TABLE A- II. Army development indicators for developing activities - Continued.**

| ARMY | |
|---|---|
| XBB | U.S. Army Electronics Command, Proc & Prd Div., Fort Huacuca AZ |
| XDD | U.S. Army Signal Air Defense Engineering Agency, fort Geroge C. Meade, MD 20755 |
| XWL | U.S. Army Limited Warfare Laboratories, Aberdeen Proving Ground, MD 21005 |
| XPM | U.S. Army Project Michigan, Ypsilanti, MI 48197 |

**TABLE A- III. Navy development indicators for developing activities.**

| NAVY | |
|---|---|
| XB | Naval Research Laboratory, Washington DC 20375 |
| XG | Naval Ocean Systems Center, San Diego, CA 92152 |
| XJ | Naval Air Development Center, Warminster, PA 18974 |
| XN | Department of the Navy, Washington DC 20360 |
| XU | Naval Underwater Systems Center, New London Laboratory, New London, CT 06320 |
| XAN | Naval Avionics Center, Indianapolis, IN 46218 |
| XCA | Naval Surface Warfare Center, Crane Division |
| XCL | Naval Air Warfare Center, Weapons Division |
| XDV | Naval Surface Warfare Center, Dahlgren Division |
| XGS | Ground Support Equipment Division, Naval Air Engineering Center, Lakehurst, NJ 08733 |
| XIH | Naval Surface Warfare Center, , Indian Head Explosives Munitions Disposal Technology Division |
| XMG | Pacific Missile Test Center, Point Mugu, CA 93042 |
| XUC | Naval Undersea Center, San Diego, CA 93132 |
| XPC | Naval Coastal Systems Laboratory, Panama City, FL 32407 |
| XWH | Naval Surface Warfare Center, Indian Head Explosives Ordinancy Disposal Technology Division Picatinny Detachment |
| Inactive | |
| XCR | Naval Weapons Center, Corona Laboratory, Corona CA 91720 |
| XZ | Bureau of Naval Weapons Activities |

**TABLE A- IV. Other Development indicators for developing activities.**

| CANADA | |
|---|---|
| XP | Canadian Department of Defense, Ottawa, Canada |



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# APPENDIX B

# PHOTOGRAPHIC CODES

## B.1 SCOPE

### B.1.1 Scope.

This appendix lists the photographic codes which are to be used to identify photographic equipment for the preparation of nomenclature or type designation requests.

## B.2 APPLICABLE DOCUMENTS

This section does not apply.

## B.3 DEFINITIONS AND ACRONYMS

This section does not apply.

## B.4 GENERAL REQUIREMENTS

### B.4.1 List of tables.

The following tables list the photographic codes which are to be used to identify photographic equipment (see Figure 1, Note 1).

a. Table B- I Photographic codes for picture taking equipment (cameras).

b. Table B- II Photographic codes for picture processing equipment.

c. Table B- III Photographic codes for picture using equipment.

**TABLE B- I. Photographic codes for picture taking equipment (cameras).**

| Category letter | Mission Letter |
|---|---|
| K - Camera.<br>L - Accessories attachments or components for cameras. | A – Reconnaissance.<br>B – Strike recording.<br><br>C – Aerial mapping.<br>D – Scope recording.<br>E – Still picture. (Not otherwise classified.)<br>F – Motion picture. (Not otherwise classified.)<br>G – Special purpose. (Including instrumentation.)<br>M – Miscellaneous. (Applying to more than one of the above types; used only with accessory letters.)<br>S – Set or system. |



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**TABLE B- II. Photographic codes for picture processing equipment.**

| Category Letter | Mission Letter |
|---|---|
| E - Processing equipment.<br>F - Accessories, attachments or components for processing equipment. | H – Processing machines.<br>J – Developers.<br><br>K – Washers.<br>L – Driers.<br>N – Printers. (Including rectifying printers, motion picture printers, contact printers and projection printers.)<br>M – Miscellaneous. (Applying to more than one of the above types; used only with accessory letters.)<br>S – Set or system. |

**TABLE B- III. Photographic codes for picture using equipment.**

| Category Letter | Mission Letter |
|---|---|
| A - Using equipment<br>B - Accessories, attachments or components for using equipment | P – Projectors (Still).<br>Q – Projectors (Motion picture).<br><br>R – Viewing devices.<br>M – Miscellaneous. (Applying to more than one of the above types; used only with accessory letters.)<br>S – Set or system. |



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# APPENDIX C

# EQUIPMENT INDICATORS

## C.1 SCOPE

### C.1.1 Scope.
This appendix lists the equipment indicators which are to be used to identify equipment for the preparation of nomenclature or type designation requests.

## C.2 APPLICABLE DOCUMENTS
This section does not apply.

## C.3 DEFINITIONS AND ACRONYMS
This section does not apply.

## C.4 GENERAL REQUIREMENTS

### C.4.1 List of tables.
The following table lists the equipment indicators which are to be used to identify equipment (see Figure 2).

a. Table C- I Equipment indicators.



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**TABLE C- I. Equipment indicators.**

| INSTALLATION (1st Indicator) | TYPE OF EQUIPMENT (2nd Indicator) | PURPOSE (3rd Indicator) | Miscellaneous Identification |
|---|---|---|---|
| A – Aerospace vehicle (Installed in or on vehicle. Not mission expendable)<br><br>B – Aerospace vehicle (Mission expendable)<br><br>C – Combination (Ground and airborne)<br><br>E – Ground, Not fixed<br><br>F – Ground, Fixed<br><br>M – Ground, Self-contained (Movable, includes vehicle but not self-propelled)<br><br>N – Aircraft or Missile transported (Not installed in or on vehicle, not mission expendable) | 22 – Apparel<br>23 – Chemical<br>24 – Electrical<br>25 – Explosive<br>26 – Gaseous<br>27 – Hydraulic<br>28 – Materials, Pliable (fabric, rubber, etc.)<br>29 – Materials, Rigid (metals, wood, etc.)<br>32 – Mechanical<br>33 – Nuclear<br>34 – Pneumatic<br>35 – Optical<br>36 – Opti-mechanical<br>37 – Electromechanical<br>38 – Invisible light (Infrared)<br>39 – Inertial<br>40 – Electro-optical<br>41 – Pneumatic-Mechanical<br>42 – Electrohydraulic<br>43 – Manual<br>44 – Internal Combustion<br>45 – Biological<br>46 – Pneumatic-hydraulic<br>47 – Electropneumatic<br>48 – Hydromechanical | A – Aerospace vehicle support<br><br>B – Bombing or fire control or both (Nonelectronic)<br><br>C – Air conditioning<br><br>D – Detection<br><br>E – Destruction<br><br>G – Flight control or navigation or both (Nonelectronic)<br><br>H – Loading and cargo handling<br><br>J – Indicating<br><br>K – Munitions handling<br><br>L – Lubricating<br><br>M – Maintenance, Aerospace vehicle | T – Training<br><br>V – Variable configuration |



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**TABLE C- I. Equipment indicators – Continued.**

| INSTALLATION (1st Indicator) | TYPE OF EQUIPMENT (2nd Indicator) | PURPOSE (3rd Indicator) | Miscellaneous Identification |
|---|---|---|---|
| P – Personal use (Held or worn by individual)<br><br>S – Ground, Self-propelled (Includes vehicle)<br><br>U – Multi-installation<br><br>W – Water (Surface or submerged) | 49 – Gunnery<br>82 – Mobile deployment (Bare base) – Miscellaneous<br>83 – Mobile deployment (Bare base) – Medical including dental, surgical, x-ray, etc.<br>84 – Mobile deployment (Bare base) – Billeting/ Administration<br>85 – Mobile deployment (Bare base) – Shop facilities – all types<br>86 – Mobile deployment (Bare base) – Food servicing including kitchen, dining, etc.<br>99 – Miscellaneous<br><br>**NOTE:** Where more than one type number applies use the one most applicable. | P – Protection<br><br>Q – Reconnaissance (Nonelectronic)<br><br>R – Refueling<br><br>S – Personnel support<br><br>T – Testing<br><br>U – Special, Not otherwise covered or combination of purposes<br><br>W – Graphic art<br><br>X – Identification<br><br>Y – Dissemination | |



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# APPENDIX D

## GROUP OR UNIT INDICATORS - BASIC PORTION

### D.1 SCOPE

**D.1.1 Scope.**
This appendix lists the group or unit indicators (alphabetically by indicator code) which are to be used to identify group or units for the preparation of nomenclature or type designation requests.

### D.2 APPLICABLE DOCUMENTS
This section does not apply.

### D.3 DEFINITIONS AND ACRONYMS
This section does not apply.

### D.4 GENERAL REQUIREMENTS

**D.4.1 List of tables.**
The following table lists the group or unit indicators for the first two of three letter combination used to identify a group or unit (see 5.2.6 and 5.2.7).

a. Table D- I Group or unit indicators – basic portion. First two of three letter combination (indicator code sequence).

**TABLE D- I. Group or unit indicators basic portion. First two of three letter combination (indicator code sequence).**

| Code | Indicator | Examples of use |
|---|---|---|
| AA | Altitude indicating instruments | Altimeter, rate of climb indicators. |
| AB | Flight instruments | Accelerometer; free air temperature indicators; aircraft clocks; and other flight not otherwise covered. |
| AC | Compressors, Air | Air compressors; excludes oxygen or other specific gaseous compressors, may or may not include prime mover. |
| AD | Adapting items | Items used to adapt one item to another. (Do not use when CD applies.) |
| AE | Engine instruments | Engine instruments not otherwise covered. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| AG | Pressure indicating instruments | Cabin pressure, hydraulic pressure, etc. (Excludes engine instruments EG.) |
| AH | Handling items | Miscellaneous items used in ground handling, pry and tow bars, etc. |
| AL | Position indicating instruments | Rudder, trim, tabs, etc. |
| AM | Ground vehicle maintenance items | Brake bonders, lifts, jacks, etc. for use on ground vehicles. |
| AP | Miscellaneous autopilot components | Any of the components that make up an autopilot. |
| AQ | Navigational instruments | Sextants, compass; any other instruments not otherwise covered. |
| AR | Attitude indicating instruments | Attitude, pitch and yaw, roll turn and slip. |
| AS | Amplifiers, Signal | Amplifiers between data source and instruments. (Not associated with communications equipment.) |
| AT | Actuating items | Devices which provide mechanical actuation for the operation of other items. |
| AV | Speed indicating instruments | Speed, Mach No., etc. |
| BA | Aircraft arresting system components | Any of the components that make up an arresting system. |
| BB | Explosive items | Such as bursters, igniters, gas generators, spotting charges, boosters, squibs, etc. (Not otherwise covered by other indicators.) |
| BC | Battery chargers | Any style battery charger. |
| BD | Bomb, Simulated | Disposable simulated bombs used for test and/or training purposes. May contain small amounts of live explosive or pyrotechnic components to add realism. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| BG | Bags | Flyers, briefcases, carrying cases, etc. |
| BL | Bomb | Any individual item of nonnuclear munitions designed to be dropped. |
| BN | Buoys | Navigation, channel marking, marking, etc. |
| BP | Assessing and briefing items, Projection | Items projecting an image to a screen; used for graphic arts purposes[1/]. |
| BR | Bomb racks and shackles | Includes all types of internal and external bomb racks and shackles. |
| BS | Munitions stabilizing and retarding devices | Includes fin assemblies, bomb retarders, and aerodynamic stabilizers. |
| BT | Batteries | Used to start vehicles and support equipment, i.e., thermal, lead acid, NiCad, etc. (Does not include primary/secondary power source for electronic equipment). |
| BV | Assessing and briefing items, Viewing | Items presenting an image directly; used for graphic arts purposes[1/]. |
| CA | Cabinets, Miscellaneous | Miscellaneous types and styles of cabinets and similar items. |
| CB | Cluster bomb | Any clustered or dispenser munition carried on internal or external aircraft bomb racks and comprised of a clustering device or munitions dispenser (SUU) and a number of aircraft bombs (BLU/BDU). |
| CC | Cartridges | Propellant or explosive device for actuating various other mechanisms. Do not use if more specific indicator exists. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| CD | Clustering devices | A clustering device containing small bombs or missiles to be dispensed from munition dispensers (SUU). May or may not be required for compatibility of bombs (BLU/BDU) in dispensers (SUU). |
| CE | Crushers, Ice | Ice crushers for crushing any type ice; dry, cube, flake, etc. |
| CF | Cartridge, Photoflash | An explosive device designed to produce a short duration light source for night photo reconnaissance missions (excludes illumination units LU). |
| CG | Cargo tiedown items | Devices used in mooring cargo, i.e., tiedowns, cables, fittings, etc. |
| CH | Cushions | Seat cushions – life-preserver and other types. |
| CJ | Cartridges, Jet engine starter | Pyrotechnic cartridges used to initiate energy to start jet engines. |
| CK | Catapult, A/C ejection seat | An item designed to propel an ejection seat with personnel from an aircraft. |
| CL | Calibrating items | Items used in calibrating operationaI equipment. |
| CM | Clothing, Miscellaneous | Miscellaneous types of clothing, such as rubber suits for bailout over water, tree jump, protection from chemical and biological agents, etc. (Other clothing not otherwise covered). |
| CN | Containers, Miscellaneous | Various containers, excludes tank type and aerial delivery containers. Includes empty containers for survival kits. |
| CP | Computing items | Computing devices, mechanical or electrical or both. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| CQ | Breathing air items | Cylinders, regulators, valves, and fittings (for breathing air only). |
| CR | Oxygen items | Cylinders, regulators, valves, fittings, etc. (for oxygen only). |
| CS | Clothing, Special | Clothing worn for protection from high altitude and "G" forces(excludes headgear, handgear, and footwear). |
| CT | Containers, Aerial delivery | Containers for aerial delivery of equipment and supplies. |
| CV | Covers | Miscellaneous types and styles of covers, heated or not heated. |
| CW | Clothing, Warmth | Clothing worn for warmth; jackets, trousers, coveralls, etc. (excludes headgear, handwear, and footwear). |
| CX | Charge, Smoke, Impact marking | A chemical substance for use in warheads, rockets, bombs, and guided missiles as a signal. When ignited smoke is produced. |
| CY | Cartridge case | Gun ammunition cartridge cases, primed or unprimed. |
| DC | Vehicle control devices | Various types of vehicle control devices. |
| DD | Demolition and destructive | Explosive devices intended for demolition and destructive purposes. Includes items designed to prevent the enemy from obtaining usefulness of captured equipment. |
| DE | Dehumidifying items | Desiccators, dehumidifiers, etc. |
| DH | Detecting items | Hydrophones, magnetometers, etc. |
| DP | Duplicating items | Dry-developing machines, copying machines, etc; used for graphic arts purposes[1/]. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| DS | Target detecting devices | Major components of a proximity fuze; an electrical or electro-optical assembly. |
| DT | Timing devices | Intervalometers, timers, nonaircraft clocks, watches, etc.[1/] |
| EA | Performance indicating instruments, Engine | Engine analyzers, cruise control indicators, etc. |
| ED | Eye protective devices (Separate items or units) | Eye protective devices which are designed for use independent of headgear or helmets. Includes those devices which can be used with or independent of headgear. Goggles, sunglasses, face shields, etc. |
| EE | Eye protective devices for headgear | Eye protective devices which are designed only for attachment onto headgear, helmets or other items covered under HG (i.e, face shields, sun shields, or goggles). |
| EF | Fuel measuring instruments | Rate-of-flow gauges and fuel remaining gauges. |
| EG | Pressure measuring instruments, Engine | Engine pressure, all types. |
| EH | Temperature indicating instruments, Engine | Engine temperature indicating instruments, all types. |
| EM | Electrical power supply items, Ground (with prime mover) | Ground electrical power supply items, generators, power plants, etc. |
| EP | Electrical power supply items, Ground (without prime mover) | Ground electrical power supply items, generators, power plants, etc. |
| ER | Tachometers, Engine | All types of engine tachometers. |
| ET | Engine transportation and handling devices | Stands, dollies,racks,etc., for engine transportation and handling during maintenance and change. |
| FD | Directional indicating instruments | Directional gyros, directional indicators, etc. |
| FE | Fire extinguishers | Any type of fire extinguisher. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| FF | Fuel filters | All types. |
| FL | Inflating equipment | For inflating liferafts and similar equipment (manual, electric, or gas powered). |
| FM | Fuzes, Munitions | Any type or style munition fuze (Excludes FS). |
| FP | Fuel tanks, Pylon-mounted | External pylon or rack-mounted tanks (jettisonable). |
| FR | Refrigerators | Mechanical refrigerators. |
| FS | Munition fuze safety – Arming device | A mechanism which prevents or allows the warhead train of explosives to operate. |
| FT | Fuel tanks, Vehicle, External | Externally mounted vehicle fuel tanks (Excludes FP). |
| FW | Footwear | Shoes, boots, mukluks, etc. |
| FZ | Munitions, Fuze-related items | Fuze related items such as delay elements, leads, arming shafts, fuze arming drive assemblies, arming wires, or lanyards, etc. |
| GA | Guns, Airborne | Applicable only to airborne guns. |
| GB | Guided bombs | Nonself-propelled vehicles with guidance systems. |
| GC | Gaseous converters, Chargers and Rechargers | For items which convert liquid to gas, charge gas containers, cartridges, etc. |
| GD | Dummy guided missiles | Dummy versions of guided missiles designed to provide training in missile assembly, handling, loading. etc. |
| GE | Generators, Electrical, Vehicle | Vehicle electrical generating equipment. |
| GF | Gun related items | Ammunition feeders, loaders, storage drums, etc. (does not include pods, tanks, or ammunition). |
| GG | Generator, Gaseous | Oxygen, C02 nitrogen generators, etc. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| GL | Engines, Ground | Engines used to power ground equipment. |
| GR | Ground refueling items | Hoses, nozzles, etc. |
| GS | Ground support items | For miscellaneous ground support items not otherwise covered. |
| GT | Weapon turrets | Any type or style weapon turret. Excludes gun pod units (GPU). |
| GU | Guns | Applicable to guns other than airborne. |
| HA | Handwear | Gloves, inserts, mittens, etc. |
| HB | Harnesses and belts, Safety | For use in restraining, etc. |
| HC | Cargo handling items | For items used in loading and handling cargo. |
| HD | Heating devices | Space, ovens, immersion, etc. |
| HG | Headgear | Helmets, caps, etc. |
| HL | Hoisting items | Lifts, slings, etc. |
| HP | Personnel aerial delivery items | Items for low level air delivery of personnel. |
| HR | Electrical harnesses and cables | Used as hookup or connection for various other equipment. |
| HS | Hose, Oxygen, Breathing | Flexible hose for connecting oxygen breathing masks (MB) to a fixed or rigid oxygen source. (Excludes bulk hose.) |
| HT | Hose, Air breathing | Flexible hose for connecting air breathing masks (MC) to a fixed or rigid breathing air source. (Excludes bulk hose.) |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| JA | Initiator, Cartridge actuated | An item designed to provide gas pressure for activation of various components such as canopy removers, thrusters, catapults, etc. |
| JE | Electrical distribution, Junction and interconnecting boxes and devices | Installed in a vehicle or used on the ground. |
| KA | Unfilled clustering device | Various types of devices to effect clusters for dropping small bombs and missiles. May or may not be required for compatibility of bombs (BLU) in dispensers (SUU). |
| KB | Cluster bomb unit, Simulated | Any simulated cluster bomb or dispenser munition, designed for test and/or training purposes. May contain live pyrotechnic components. |
| KD | Clustering devices, Simulated | Any clustered or dispenser munition, designed for test and/or training purposes only. May contain live pyrotechnic components. |
| KM | Kits, Miscellaneous | For kits not designatable as sets, groups, or units. |
| LA | Launching mechanisms, Vehicle installed | Vehicle installed launchers. (Do not use if SU or PW applies.) |
| LB | Lubricating devices | Lubricators, lubrication spray guns, etc. |
| LF | Lighting items, Ground | Floodlights, spotlights, extensions, explosion-proof inspection lights. |
| LK | Links for ammunition | For carrying or feeding arnmunition to automatic weapons. |
| LM | Launching mechanisms, Ground | Ground launchers for rockets, missiles, targets. |
| LN | Lights (Navigation) | For navigaition in mine counter measures missions. May be flashing or steady beacon. |
| LP | Life preservers | Any style or size life preserver. |
| LR | Liferafts | Any style or size liferaft. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| LT | Vehicle installed lights | Lights installed in or on vehicles. |
| LU | Illumination unit | Air munitions items such as flares, searchlights, or other visible light devices used to provide illumination of battle areas and photo reconnaissance targets. (Also includes signal flares.) |
| LW | Laser weapons | High power lasers used to destroy materiel. (Not to include laser used to designate targets.) |
| MA | Miscellaneous armament items | Not otherwise covered. (Excludes bomb racks and shackles.) |
| MB | Masks, Breathing oxygen | Used only for breathing oxygen masks. |
| MC | Masks, Breathing air | Used only for breathing air face masks. (Excludes oxygen masks.) |
| MD | Miscellaneous munitions, Simulated | Simulated practice munitions, where the simulated live munition is not identified in the designation system. |
| MH | Munitions handling items | Not otherwise covered, dollies, slings, etc. |
| MJ | Munition-countermeasures | Any loaded dispenser munition-type device dropped from vehicle for the purpose of confusing or misleading enemy forces. |
| ML | Miscellaneous munitions, Live | Live munitions not otherwise provided for. |
| MM | Vehicle maintenance items | Not otherwise covered, i.e., vehicle jacks, tools, etc. |
| MP | Masks, Protective | For protective face masks, other than breathing air or oxygen, i.e., tree jump, environmental protective, etc. |
| MS | Maintenance stands | Stands, platforms, etc. |
| MT | Mounts | Any style mount. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| MX | Miscellaneous | Items not otherwise covered. |
| NC | Non-stereo, Charting items | Chart and map compilation items utilizing principles other than stereoscopic projection and viewing, used for graphic arts purposes[1/]. |
| NM | Nuclear measuring items | Radiometers, geiger counters, etc. |
| NR | Miscellaneous nitrogen items | Includes cylinders, regulators, valves, and associated items for nitrogen use only. |
| PA | Munitions dispensing devices, External | External pods, tanks, clip-ins, or other devices for dispensing munitions, launching rockets, dispensing B-C materials - mounted external of vehicle. (Excludes SU.) |
| PC | Parachute components | Canopies, harnesses, packs, pilot chutes, etc. (Complete parachute assembly its designated by an equipment-designation.) |
| PD | Printed data for dispensing | Prepackaged leaflets and miscellaneous printed data dispensed directly from vehicle. |
| PF | Panel, Electrical | Control panels, etc. |
| PF | Propellent control items | Used to identify components utilized in rocket propellent flow control. |
| PG | Gun ammunition | All types of gun ammunition. |
| PJ | Projectiles | Projectiles, including as appropriate, fuzing, explosive, incendiary, penetration, tracer, etc. |
| PL | Plotting devices | Table, scales, charts, holders, etc.[1/] |
| PM | Pumps | Various types of pumps, manual or electric. |
| PP | Printing items | Printing presses, etc.; used for graphic arts purposes[1/]. |
| PR | Protective items | Armor, vests, kits, etc. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| PT | Power units, Auxiliary | Auxiliary power units that operate aircraft equipment while on the ground. |
| PV | Primer | Percussion or electrically initiated component of cartridge for initiation of propellant. |
| PW | Munitions dispensing devices | Nonexpendable empty pods, tanks, clip-ins, or other devices for dispensing munitions, launching rockets, dispensing B-C materials-mounted inside of vehicle. (Excludes SU.) |
| RA | Remover, Canopy | An item designed to jettision the canopy from an aerospace vehicle to provide an exit for personnel. |
| RB | Rocket and launcher units | A combination of a launcher (LAU) and conventional munitions type rockets (RLU) that is carried and functioned from external stations. Excludes cluster bomb units (CBU). |
| RD | Rockets, Simulated | A self-propelled, noncontrollable (after firing) rocket (dummy or practice) used for test and training purposes. |
| RE | Rescue devices | Baskets, towing ropes, etc. |
| RF | In-flight refueling items | Nozzles, hose, etc., used for in-flight refueling. |
| RG | Regulating items, Electrical items | Devices used to regulate electrical current, voltage, frequency, etc. |
| RH | Reconnaissance interpretation | Film plotting table, stereoscopes, magnifier, etc., used for graphic arts purposes[1/]. |
| RL | Rockets, Live (Complete round) | A self-propelled, noncontrollable (after firing) conventional munitions type rocket. |
| RM | Reels and reeling mechanisms | Mechanisms used for reeling cable, towrope, wire, etc., in or out. (Excludes the cable, towrope, wire, etc., see TQ.) |
| RV | Re-entry vehicle (Dummy) | Dummy or practice types of re-entry vehicles. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| RW | Re-entry vehicle | Actual re-entry vehicle, that portion of a space vehicle designed to re-enter the earth's atmosphere in the terminal portion of its trajectory. |
| RY | Relays, Electrical | Any electrical relay. |
| SA | Sights | All types of optical bomb, gun, and rocket sights. |
| SB | Stabilizing items | Gyroscopes, etc. |
| SC | Stereo-Charting items | Items used in compiling charts and maps utilizing the principles of stereoscopic projection and viewing; used used for graphic arts purposes[1/]. |
| SD | Signal devices | Signal lights, hand held and otherwise warning devices including vehicle installed. |
| SE | Seats | Seats, pilots, crew members, and passengers. (Excludes ejection seats.) |
| SG | Starters, Ground | Starting units used on the ground to start engines. (Not installed on the engine being started). |
| SH | Shelters | Structures used for maintenance, storage, and miscellaneous uses. |
| SJ | Seats, Ejection | Pilots and crew members ejection seats. |
| SK | Seat kits | Rigid or soft type for integration into ejection seats. (Complete kits only; for empty containers only see CN.) |
| SM | Simulators | Items which replicate the operation of real equipment. |
| SP | Sweeping items (Mines) | Floats, otters, depressors, cutters, etc. |
| SR | Survival items | Sustenance kits and components of sustenance kits, knives, etc. |
| ST | Starters, Installed | Any type of starter installed on engine. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| SU | Suspension and release unit Stores (Munitions dispenser) | An item carried internally or externally on bomb racks and used to transport and deliver and/or function smaller nonnuclear munitions items to include bombs and guns. (Also a CBU less BLUs.) |
| SV | Servo-mechanisms and components | Servos, drum and brackets, motor and drives, etc. |
| SW | Switches, Electrical | Various electrical switches and similar items. |
| SX | Scoring devices | Items used in determining miss distances and firing error indication. Normally installed in targets. |
| TA | Training aids | A visual training aid. Does not normally provide "hands-on" training for students. |
| TC | Thruster, Cartridge actuated | An item operated by gases generated by firing of a cartridge. Provides thrust for opening or closing latches, hatches, etc. |
| TD | Target devices | Various types and styles of targets, tow, winged, etc. (Excludes drones.) |
| TE | Transformers, Electrical | Any size or style transformer associated with vehicle or support equipment. |
| TH | Target handling items | Cables, cutters, etc. |
| TK | Tracking equipment | Target tracking, nonelectronic. |
| TL | Tools | Tools not otherwise covered. |
| TM | Tanks, Miscellaneous | Oxygen, fuel, sampling, etc. (Excludes vehicle fuel tanks.) |
| TQ | Miscellaneous towing devices | Tow cables, wire, and other items related to towing components and devices. Do not use when RM is more applicable. |
| TR | Transmitters, Instrument | Signal transmission to instruments, rate-of-flow, compass, pressure, etc. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| TT | Test items | Any test itenn not otherwise covered. |
| TW | Tape units | Preprogrammed with operational test and checkout data. (Excludes magnetic tape units.) |
| VA | Valves | Various types of valves. Do not use where CQ and CR apply. |
| VC | Chassis, Vehicle | All types of vehicle chassis. (Complete vehicle is designated by an eqiuipment designation.) |
| VS | Viewing devices | Viewing items such as bineculars, telescopes, periscopes, etc. (Excludes bomb-gun-rocket sights SA.) |
| WA | Warhead section | An item consisting of a portion of the outer aerodynamic cast (shell or skin) of a guided missile, rocket, or like vehicle, plus a warhead, inert or loaded, which is removable from the outer case of the vehicle. Do not use if WD, WE, and WT apply[2/]. |
| WB | Body section | Items which comprise a portion of the outer aerodynamic shell or skin of a guided missile, or other vehicle, and when used in conjunction with other sections (warhead sections, propulsion sections, and the like) forms a complete guided missile or other vehicle. Do not use when WA, WG, WN, or WP apply. |
| WC | Control section | Items which comprise a portion of the outer aerodynamic shell or skin of a guided missile, or other vehicle. It includes the items (actuators, internal power sources, umbilical connectors, and the like) necessary to effect flight control and stability. Do not use if WA, WG, WP, etc., apply. |



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**TABLE D- I. Group or unit indicators - basic portion. First two of three letter combination (indicator code sequence) - Continued.**

| Code | Indicator | Examples of use |
|---|---|---|
| WD | Warhead, Explosive | An explosive loaded item dcsigned to be mounted in or on a guided missile, rocket, or the like. The configuration may form a portion of the outer case of the delivery vehicle when this portion of the case is designed not to be removed from its contents. Excludes WA, WE, WD, and WT[2/]. |
| WE | Warhead, Empty | Empty warheads. |
| WF | Warhead | Any warhead other than explosive, i.e., penetrator. Excludes WA, WE, and WT[2/]. |
| WG | Guidance section | Items which comprise a portion of the outer aerodynamic shell or skin of a guided missile, and which includes all or a major portion of the necessary electrical and/or electronic equipment or items used to effect guidance of the missile in flight. |
| WM | Weather devices (Nucleating) | Used for atmospheric application; such as catalyst generators, etc. |
| WN | Nose section | Items which for the extreme forward aerodynamic portion of a guided missile, or other vehicle; may contain instrumentation, spotting charges, and/or fuzing or arming devices and the like. Do not use when WA, WC, or WG apply. (Excludes WD, WE, and WT.) |
| WP | Propulsion section | Items which comprise a portion of the outer aerodynamic shell or skin of a guided missile or other vehicle. It contains, or is designed to contain one or more engines or motors. May include movable control surfaces, stabilizing fins, or a combination of these and other items. |
| WT | Warhead, Training dummy, or Practice | Dummy, practice, and training warheads. Excludes WA, WD and WE. (For nonnuclear items, follow guidance contained in section 6.) |

1/ Not to be used if photographic indicators are applicable.
2/ Not applicable where another DoD identification system applies.



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# APPENDIX E

# SYMBOLS FOR PROPULSION ENGINES

## E.1 SCOPE

### E.1.1 Scope.

This appendix lists the symbols which are to be used to identify engine types for the preparation of nomenclature or type designation requests.

## E.2 APPLICABLE DOCUMENTS

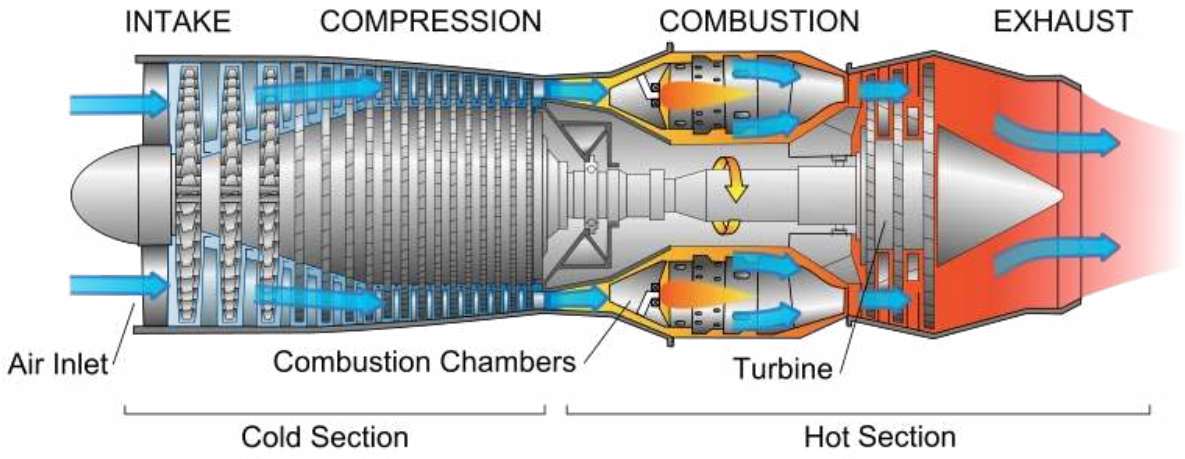
Not applicable to this section.

## E.3 DEFINITIONS AND ACRONYMS

### E.3.1 J – Turbojet.

A **turbojet** engine (see Figure E- 1) is a jet engine having a turbine-driven air compressor for supplying compressed air to the combustion chamber and a discharge nozzle for directing the heated exhaust gasses rearward. Turbojet engines produce thrust only by means of a high-speed jet of hot gasses. ***(The Engine Handbook, 1991)***

Turbojets are quite inefficient (if flown below about Mach 2) and very noisy. Most modern aircraft use turbofans instead for economic reasons. Turbojets are still very common in medium range ***cruise missiles***, due to their high speed, low frontal area and relative simplicity. ***(http://en.wikipedia.org/wiki/Turbojet)***




***(http://defencyclopedia.com/2015/05/13/explained-how-jet-engines-work/)***

**FIGURE E- 1. Typical turbojet engine.**

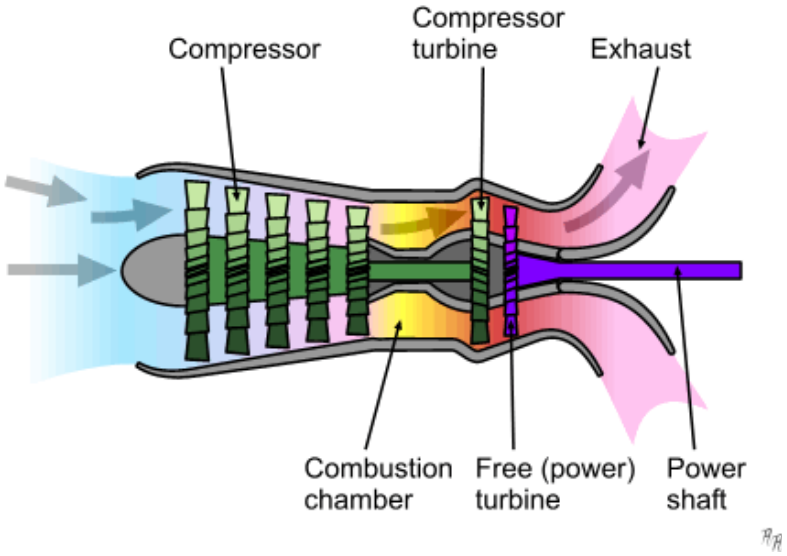


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**E.3.2 T – Turboshaft / turboprop.**

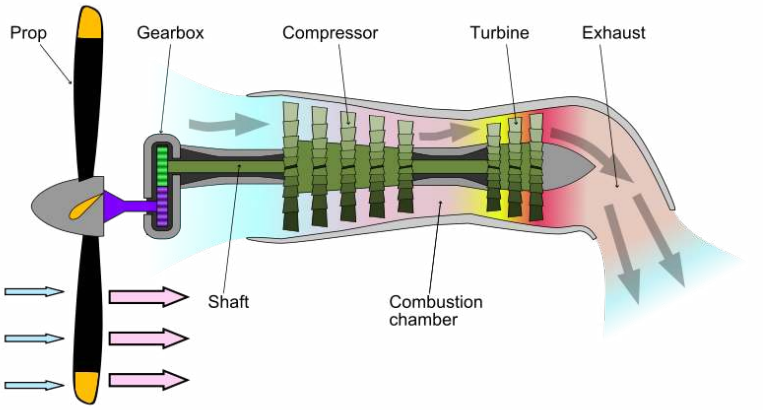
**Turboshaft** and **turboprop** engines (see Figure E- 2) are turbojet engines from which power is extracted through a shaft connected to a turbine. The compressor-drive and shaft-drive turbines may be independent or connected. Energy remaining in the gasses after they exit the turbine may be converted into jet thrust (i.e. the T56). ***(The Engine Handbook, 1991)***

**TURBOSHAFT**




*(http://defencyclopedia.com/2015/05/13/explained-how-jet-engines-work/)*

**TURBOPROP**



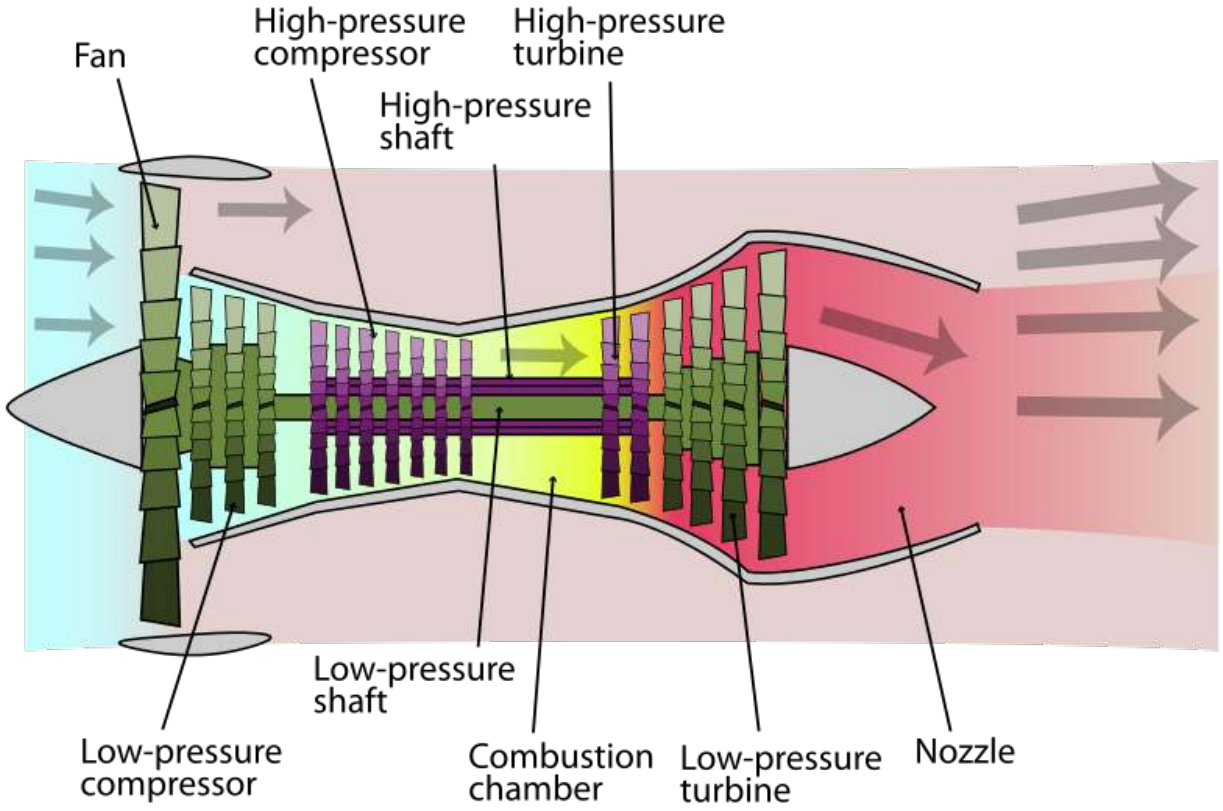

*(http://defencyclopedia.com/2015/05/13/explained-how-jet-engines-work/)*

**FIGURE E- 2. Typical turboshaft / turboprop engines.**



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**E.3.3 F – Turbofan.**
Turbofan engines (see Figure E- 3) are turbojet engines that have a turbine-driven axial-flow fan or enclosed propeller. Turbofan engines are sometimes referred to as bypass engines, since some or much of the air exiting the fan bypasses the turbojet part of the engine. High-bypass turbofans, such as TF39, produce most of their thrust through the fan. The fan may be mounted either on the rear of the engine, or the front (this is the most common configuration today). The air flowing from the fan may be ducted to the rear of the engine and mixed with the exhaust gas from the turbojet. If additional fuel is added to this combined exhaust, the result is an augmented turbofan (i.e. the F100). ***(The Engine Handbook, 1991)***




***(http://defencyclopedia.com/2015/05/13/explained-how-jet-engines-work/)***

**FIGURE E- 3. Typical turbofan engine.**

**E.3.4 Reciprocating.**
Reciprocating aircraft engines (see Figure E- 4) work on the same principle as the engine in automobiles. Fuel is burned in contained combustion chambers (the cylinders) and mechanical energy is extracted from the resulting hot gasses pushing against pistons. The pistons are connected to a shaft, which then may be used to turn a propeller or other device. Aircraft reciprocating engines may have their cylinders and pistons mounted inline (like most automobile engines), in opposition (like the Volkswagen Beetle), or arranged radially (in a circle). ***(The Engine Handbook, 1991)***



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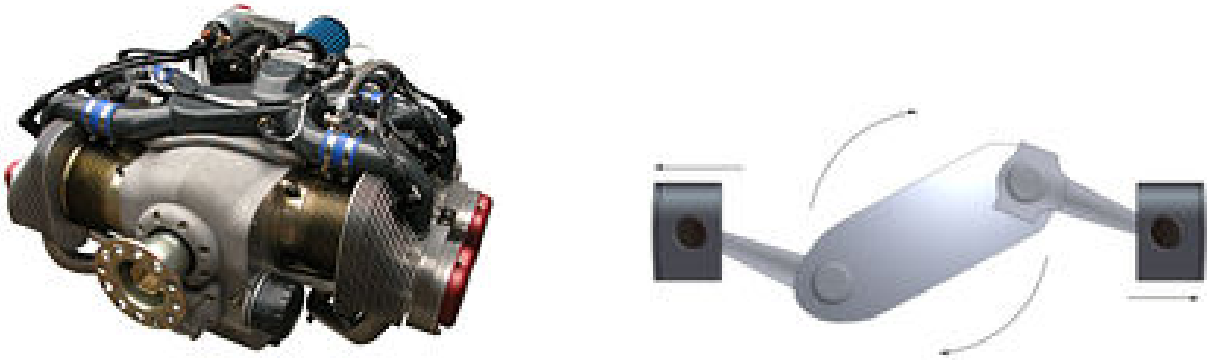
(*http://www.webpages.uidaho.edu/mindworks/Solidworks/past_projects.htm*)

**FIGURE E- 4. Reciprocating engine.**

**E.3.4.1 O – Opposed (boxer).**
A flat engine or horizontally-**opposed** cylinder engine (see Figure E- 5) is an internal combustion engine with pistons that are all relatively horizontal. A straight engine canted 90 degrees from straight up is a flat engine, as is one in which the cylinders are arranged in two banks on either side of a single crankshaft. In both configurations, the motion of all the pistons is in the horizontal plane.



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*(http://en.wikipedia.org/wiki/Flat_engine)*

**FIGURE E- 5. Typical opposed (boxer) engine.**

**E.3.4.2 R – Radial.**
The radial engine (see Figure E- 6) is an internal combustion engine configuration in which the cylinders point outward from a central crankshaft like the spokes on a wheel. This configuration was very commonly used in aircraft engines before being superseded by turboshaft and turbojet engines. It is a reciprocating engine (piston).

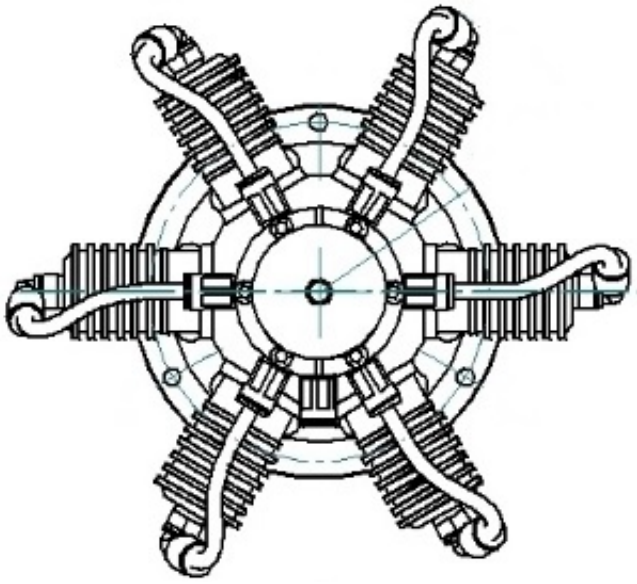


**FIGURE E- 6. Typical radial engine.**



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**E.3.4.3 V – Vee (v-engine).**
A “V”-engine (see Figure E- 7) is a common configuration for an internal combustion engine. The cylinders and pistons are aligned, in two separate planes, so that they appear to be in a **"V"** when viewed along the axis of the crankshaft. Certain types of V engine have been built as inverted engines, most commonly for aircraft. Advantages include better visibility in a single-engined airplane, and lower centre of gravity.

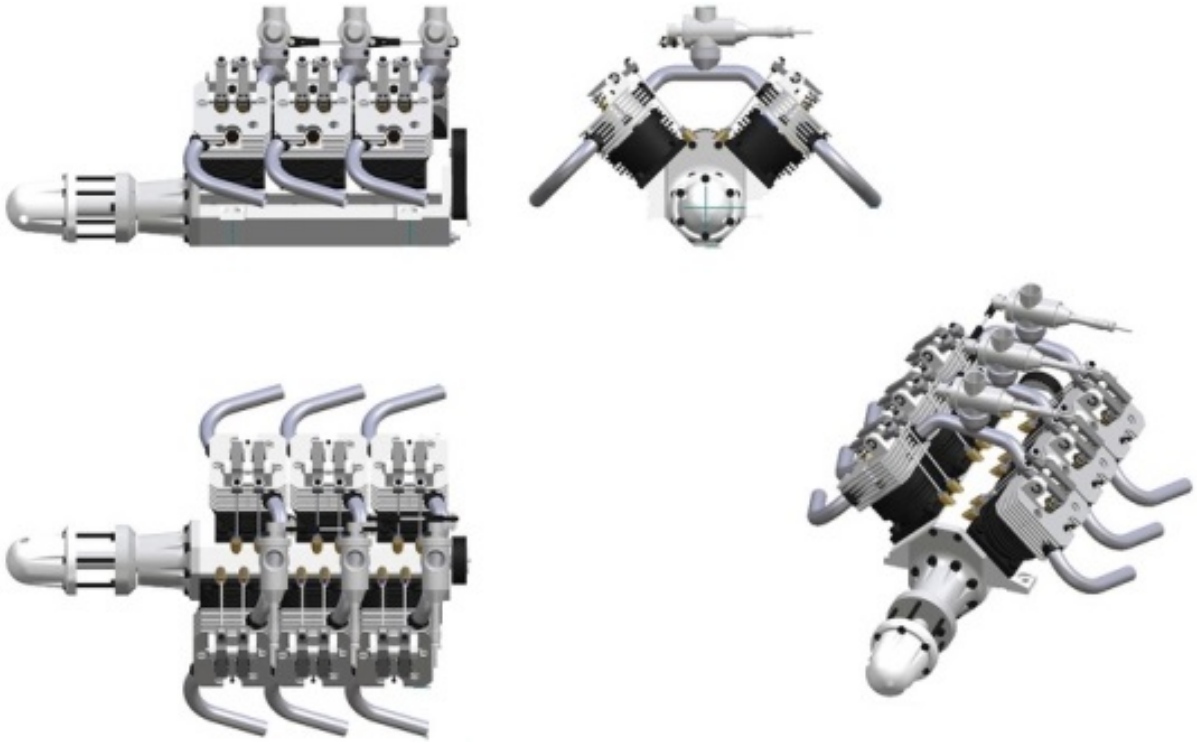
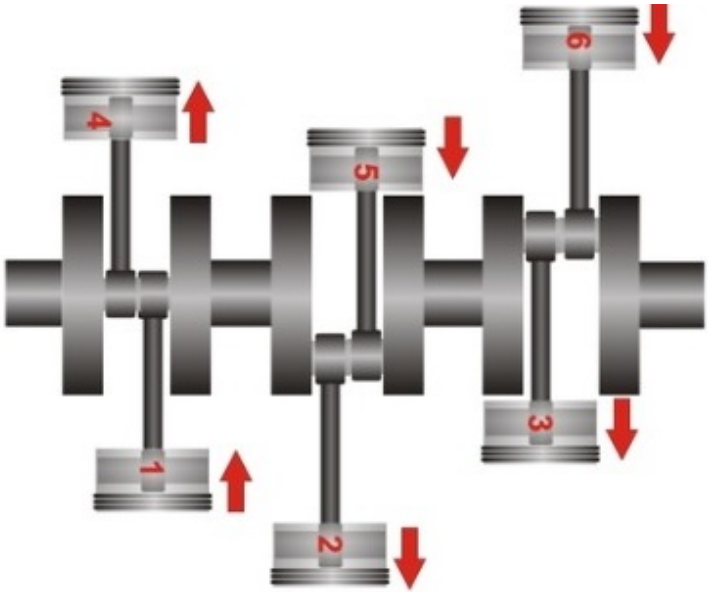


**FIGURE E- 7. “V”-engine.**



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**E.3.4.3.1 Flat-V – Flat Vee (v-engine).**
The "Flat-Vee"-engine (see Figure E- 8) has cylinder and piston positions in a relatively single horizontal plane (cylinder positions at 0 and 180 degrees). This is similar to a horizontally opposed / Boxer engine (see Figure E- 5), since both engine types have cylinder positions at or near 0 and 180 degrees. What makes these two differ is the use or non-use of the same crankpin for the two horizontally opposed pistons. The Flat-Vee shares the same crankpin for two horizontally opposed pistons (note the direction of the piston travel on Figure E- 8, due to the sharing of the crankpin), whereas the Boxer type does not share, but uses two different crankpins for the two horizontally opposed pistons. Examples include World War II German engines produced by Daimler-Benz and Jumo. ***(http://en.wikipedia.org/wiki/V_engine)***



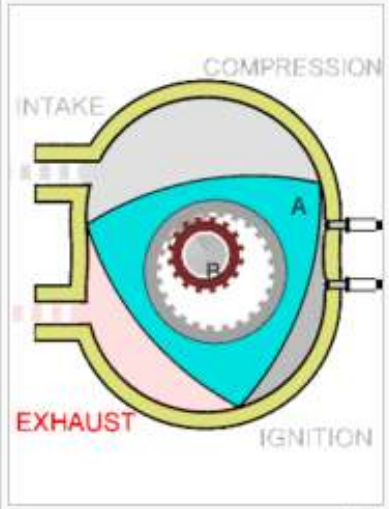

***(http://en.wikipedia.org/wiki/Flat_engine)***

**FIGURE E- 8. "Flat-Vee"-engine.**

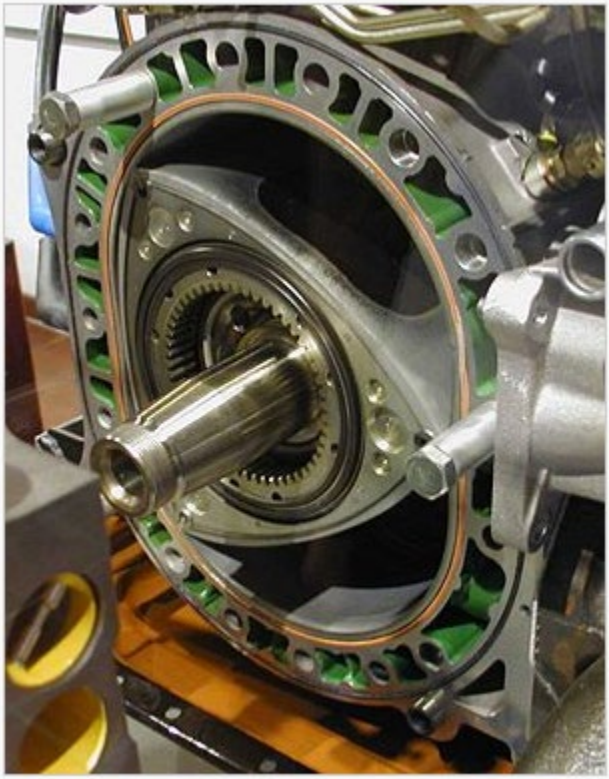
**E.3.4.4C – Rotating combustion (rotary / wankel).**
A rotating combustion engine (see Figure E- 9) is an internal combustion engine which uses a rotary design to convert pressure into a rotating motion instead of using reciprocating pistons. Its four-stroke cycle is generally generated in a space between the inside of an oval-like epitrochoid-shaped housing (see Figure E- 10) and a roughly triangular rotor. This design delivers smooth high-rpm power from a compact, lightweight engine.



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The Wankel cycle. The "A" marks one of the three apexes of the rotor. The "B" marks the eccentric shaft and the white portion is the lobe of the eccentric shaft. The shaft turns three times for each rotation of the rotor around the lobe and once for each orbital revolution around the eccentric shaft.



A Wankel engine in Deutsches Museum Munich, Germany.

*(http://en.wikipedia.org/wiki/Wankel_engine)*

**FIGURE E- 9. Rotating combustion (rotary / wankel) engine.**

## Epitrochoid

From Wikipedia, the free encyclopedia

An **epitrochoid** (IPA: /ɛpɨˈtrɒkɔɪd, -ˈtroʊ-/) is a roulette traced by a point attached to a circle of radius $r$ rolling around the outside of a fixed circle of radius $R$, where the point is a distance $d$ from the center of the exterior circle.

The parametric equations for an epitrochoid are

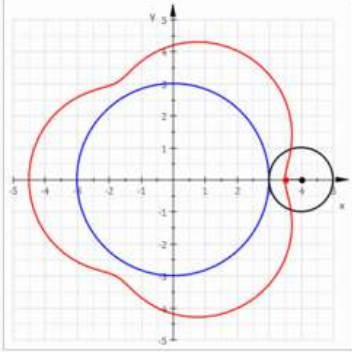
$$x = (R+r)\cos\theta - d\cos\left(\frac{R+r}{r}\theta\right),$$
$$y = (R+r)\sin\theta - d\sin\left(\frac{R+r}{r}\theta\right).$$

Special cases include the limaçon with $R = r$ and the epicycloid with $d = r$.

The classic Spirograph toy traces out epitrochoid and hypotrochoid curves.

The orbits of planets in the once popular geocentric Ptolemaic system are epitrochoids.

The stator of the Wankel engine is an epitrochoid.



The epitrochoid with $R = 3$, $r = 1$ and $d = 1/2$

*(http://en.wikipedia.org/wiki/Epitrochoid)*

**FIGURE E- 10. Epitrochoid-shaped housing.**



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### E.3.5 A – Adaptive engine.

The adaptive engine (see Figure E- 11) architecture is a revolutionary advancement in turbine engine capability from a baseline turbofan design. The components of an adaptive engine still maintain the compression, combustion, and expansion techniques of traditional turbofan engines with fan, compressor, combustor, turbine, and exhaust nozzle components. The adaptive engine features include the addition of a third stream of bypass cooling air that enables improved fuel consumption, higher thrust, and greater thermal capacity while maintaining full-life durability and sustainability. Additionally, the adaptive engine architecture provides performance matching across the flight envelope to optimize between thrust and fuel efficiency depending on the mission requirements.

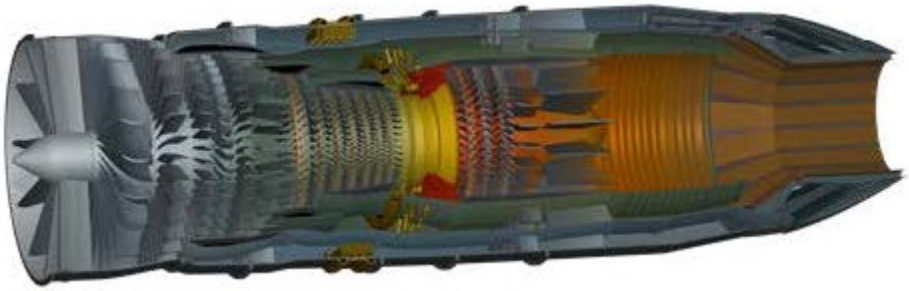


**FIGURE E- 11. Adaptive engine.**

## E.4 GENERAL REQUIREMENTS

### E.4.1 List of tables.

The following tables list the symbols, which are to be used to identify air breathing and non-air breathing engines (see Figures 3 and 4).

a. Table E-I Type letter symbols for air breathing engines.
b. Table E-II Type letter symbols for non-air breathing engines.
c. Table E-III Manufacturer's and contractor's symbols for engine types.

**TABLE E- I. Type letter symbols for air breathing engines.**

| Letter Symbols | Engine Type |
|---|---|
| J | Turbojet |
| T | Turboshaft, Turboprop |
| F | Turbofan |
| O | Opposing |
| R | Radial |
| V | Vee |
| C | Rotating Combustion |



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**TABLE E- I. Type letter symbols for air breathing engines - Continued.**

| Letter Symbols | Engine Type |
|---|---|
| A | Adaptive |
| P | Any type not listed |

**TABLE E- II. Letter symbols for non air breathing engines.**

| Letter symbol | |
|---|---|
| LR | Liquid propellant rocket engine |
| SR | Solid propellant rocket motor |
| LSR | Hybrid propellant rocket engine (liquid-solid) |

**TABLE E- III. Manufacturer's and contractor's symbols.**

| Manufacturer's and Contractor's Name | Letter Symbol |
|---|---|
| Aerodyne Corp. | AE |
| Aerojet-General Corp. | AJ |
| Aeronutronics Corp. | AN |
| AiResearch Division, Garrett Corporation | GA |
| Allegany Ballistics Lab. | AB |
| Allison Div., General Motors Corp. | AD |
| AMOCO Chemicals Corp. | AC |
| Astropower, Inc. | AI |
| Atrosystems, Inc. | AS |
| Atlantic Research Corp. | AR |
| Bell Aerosystems Co. | BA |
| B.F. Goodrich Co. | BG |
| Continental Aviation and Engineering Corporation | CA |
| Curtiss-Wright Corp. | CW |
| General Atomics Aeronautical Systems, Inc. | GT |
| General Electric Co. | GE |
| Hercules Powder Co. | HP |
| Hughes Tool Co. | HT |
| Lockheed Propulsion Co. | IT |
| Marquardt Corporation | MA |
| Microturbo | MT |
| Naval Missile Center | NM |
| Naval Weapons Center | NW |
| Naval Propellant Plant No. 5 | NP |
| North American Rockwell Corp. (Rocketdyne) | NA |
| Northrup Carolina Corp. | NC |



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**TABLE E- III. Manufacturer's and contractor's symbols - Continued.**

| Manufacturer's and Contractor's Name | Letter Symbol |
|---|---|
| Olin Mathieson Co. | OM |
| Picatinny Arsenal | PA |
| Pratt & Whitney Acft Div., United Aircraft Corp. | PW |
| Reaction Motors Div., Thiokol Chemical Corp. | PM |
| Rocketdyne (now Rockwell/Hercules) | RD |
| Rocket Power, Inc. | RP |
| Rocket Research Corp. | RC |
| Rolls Royce, Lid | RR |
| Rotax | RO |
| Sunstrand Power Systems | SP |
| Thiokol Chemical Corp. | TC |
| TRW Systems | TR |
| United Aircraft a[ Canada, Ltd. | CP |
| United Aircraft of West Virginia | WV |
| United Technology Center | UT |
| Williams Research Corporation | WR |
| Any contractor not yet listed or development where contractor is not identified. | ZZ[1/] |

**1/ NOTE:** If this code is used, the designation must be revised when the contractor's code is developed or the contractor is selected.



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## CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 11

Preparing activity:
Air Force - 11

(Project SESS-2015-074)

Review activities:
Navy - SH

International interest (see 6.5)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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